

## Article

# Unified Multi-Level Grape Counting Framework with Low Annotation Cost for Natural Orchard Scenes

Xiaofeng Xie , Jingying Yang  and Zhaoli Shen \*

College of Science, Sichuan Agricultural University, Ya'an 625014, China; xfxiesicau@163.com (X.X.); jyyang51@163.com (J.Y.)

\* Correspondence: szlxiaoyao@163.com

## Abstract

Accurate multi-level grape counting in natural orchard images provides useful visual information for image-based crop-load characterization and vineyard management. However, cluster overlap, occlusion, boundary adhesion, and illumination variation complicate berry localization and cluster-wise counting. Existing detection- and density-regression-based methods remain susceptible to missed detections, false positives, and cross-cluster mis-assignment, whereas approaches relying on pixel- or instance-level supervision typically require costly annotations. To address these challenges, this paper proposes UniGC, a unified framework for multi-level grape counting. At the berry level, UniGC employs CSD-P2PNet, a P2PNet-based point-regression branch equipped with a Condition-Structure Decoupled Frequency Enhancement Block (CSD-Block), to improve the localization and counting of image-visible berries under complex illumination and occlusion using only berry point annotations. At the cluster level, a You Only Look Once (YOLO) detector predicts grape-cluster bounding boxes, which are used as prompts for the Segment Anything Model (SAM) to generate cluster instance masks. We further propose Boundary-Depth Guided Mask-Point Assignment (BDMPA), which assigns each predicted berry point to at most one grape-cluster instance and thereby produces cluster-wise berry counts. Selected key experiments on the mixed dataset organized from WGSD and GBISC were repeated using three random seeds. CSD-P2PNet achieved an MAE of  $23.4407 \pm 0.1577$ , an RMSE of  $32.8898 \pm 1.2682$ , and a point-localization F1 score of  $0.8374 \pm 0.0097$ . YOLO12-S achieved an F1 score of  $0.8433 \pm 0.0056$  and an AP50 of  $0.8789 \pm 0.0080$  for grape-cluster detection. For BpGC counting, UniGC obtained a matched-cluster MRD of  $0.1288 \pm 0.0043$ , a 1 – FVU of  $0.8589 \pm 0.0098$ , and an all-cluster penalized mean relative deviation (ACP-MRD) of  $0.5062 \pm 0.0124$ . These results demonstrate that UniGC achieves competitive multi-level grape counting performance while requiring only point and bounding-box annotations for training.



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**Keywords:** orchard intelligent counting; grape cluster detection; point regression; SAM segmentation; UniGC framework; condition-structure decoupling; mask-point assignment

## 1. Introduction

In natural orchard images, grape cluster-level detection and counting, image-visible berry-level counting, and berries-per-grape-cluster (BpGC) counting provide complementary visual information for image-based crop-load characterization, cultivation management, and precision operations. Previous studies have applied computer vision techniques

to grape yield estimation, harvesting localization, cluster-level recognition, and berry phenotypic analysis, providing a feasible way to reduce reliance on manual counting and improve field management efficiency [1–8]. Compared with whole-image or cluster-level counting alone, multi-level counting provides more structured information, including the spatial distribution of grape clusters, the number of image-visible berries, and the visible berry load within individual clusters. Such information is therefore more consistent with practical needs in fruit thinning, crop-load regulation, and precision management [9–12]. Automatically obtaining these multi-level visual measurements using computer vision has therefore become an important research direction in intelligent grape perception and precision management.

Existing studies on grape detection and counting can be broadly divided into three categories according to task level and technical approach. The first category focuses on grape cluster detection or segmentation, usually using object detection, semantic segmentation, or instance segmentation models to obtain cluster-level geometric information for tasks such as cluster counting, harvesting localization, or yield estimation [13–24]. The second category focuses on berry-level detection, segmentation, or counting, including instance segmentation and small-object detection methods, as well as point-supervised, density-map-based, or semi-supervised counting methods, to support quantitative assessment in densely distributed berry scenarios [7,11,12,25–29]. The third category further attempts to model grape cluster detection, grape berry counting, and BpGC counting as interrelated subtasks [9–12,30]. Overall, existing studies have laid an important foundation for multi-level visual analysis of grapes, but most methods still focus on single-level tasks or do not sufficiently address consistent associations among results at different levels. Therefore, establishing a reliable and efficient unified framework for multi-level grape counting requires further investigation.

Specifically, in natural orchard images, grape-cluster overlap, severe occlusion, boundary adhesion, and complex illumination variations jointly affect cluster-level detection, image-visible berry-level counting, and BpGC counting [31]. First, complex illumination variations can alter local berry texture, edge responses, and contrast distributions, making it difficult for detection- or density-regression-based methods to distinguish berry-related structures from background disturbances and thereby increasing missed and false berry detections. Second, severe occlusion, inter-cluster overlap, and boundary adhesion can weaken the separability of grape-cluster regions, making it difficult for cluster-level detection or segmentation results to provide a reliable geometric basis for subsequent counting and assignment. Furthermore, when berry-point localization results are assigned to specific grape clusters, localization errors and regional boundary uncertainties can propagate through the task pipeline, increasing the risk of cross-cluster misassignment. If pixel-level segmentation or more fine-grained instance-level supervision is used to alleviate these problems, the annotation burden can increase substantially, limiting the scalable application of such methods in real agricultural scenarios.

To address the above problems, this paper proposes UniGC, a unified framework for multi-level grape counting. UniGC follows a unified pipeline that combines berry-level point-regression counting, cluster-level detection and mask generation, and berry point-to-cluster assignment-based counting. It jointly performs image-visible berry-level counting, grape cluster-level detection and counting, and BpGC counting, while the trainable branches require only berry point annotations and grape-cluster bounding-box annotations. By using predicted grape-cluster bounding boxes as prompts for mask generation, UniGC maintains a unified multi-level inference process without requiring grape-cluster instance-mask annotations for task-specific training.

In the berry-level branch, P2PNet is used as the point-regression counting basis [32], and the Condition-Structure Decoupled Frequency Enhancement Block (CSD-Block) is introduced. This module models intermediate features from a frequency-inspired perspective. Through learnable low-pass filtering and residual decomposition, the module promotes the spatial-domain disentanglement of slowly varying condition-related components from residual structure-sensitive components. It then performs condition-decoupled enhancement and controlled recoupling on the residual structural features, thereby improving the robustness of image-visible berry-point localization under complex illumination and occlusion.

In the cluster-level branch, You Only Look Once (YOLO) is used to predict grape-cluster bounding boxes, which serve as prompts for the Segment Anything Model (SAM) to generate the corresponding cluster-level instance masks and provide geometric priors for subsequent assignment inference [33]. On this basis, the Boundary-Depth Guided Mask-Point Assignment (BDMPA) is further proposed to establish a unique point-to-cluster assignment for each predicted berry point. In addition, at the data level, the Wine Grape Instance Segmentation Dataset (WGISD) and the Sichuan Grape Berry Instance Segmentation and Counting Dataset (GBISC) [13,27] are uniformly organized and integrated to construct a data organization scheme suitable for multi-level grape detection and counting tasks. Experimental results show that, under the evaluated mixed-dataset setting, UniGC achieves competitive performance in image-visible berry counting, grape-cluster detection, and BpGC counting, while its trainable branches use only berry point annotations and grape-cluster bounding-box annotations.

The main contributions of this paper are as follows:

1. We propose UniGC, a unified framework for multi-level grape counting that supports grape berry-level counting, grape cluster-level detection and counting, and berries-per-grape-cluster (BpGC) counting.
2. We propose the Condition-Structure Decoupled Frequency Enhancement Block (CSD-Block), which improves fine-grained structural representation related to berry point localization under complex imaging conditions through learnable low-pass decomposition, structural component enhancement, and controlled recoupling.
3. We propose the Boundary-Depth Guided Mask-Point Assignment (BDMPA), which provides stable per-point assignment of berry points to grape clusters without introducing additional trainable parameters.
4. Extensive experiments verify the effectiveness of the proposed UniGC in multi-level grape counting. The results show that the framework achieves competitive performance in berry-level counting, cluster-level detection and counting, and BpGC counting.

## 2. Related Work

### 2.1. Grape Cluster-Level Detection and Segmentation

Grape cluster-level detection and segmentation aim to obtain the locations, counts, and spatial regions of grape clusters, providing key visual information for yield estimation, harvesting localization, and field management. Santos et al. [13] proposed the WGISD dataset and used deep neural networks for grape cluster detection, segmentation, and tracking, providing an important data resource for grape cluster-level visual research. Subsequently, the YOLO series of detectors has been widely used for grape cluster recognition in natural scenes [17–24]. Zhang et al. [14] achieved real-time grape cluster detection in complex natural scenes using YOLOv5s, whereas Zhang et al. [2] proposed YOLOv5-GAP for grape cluster recognition and picking-point localization under rachis occlusion.

In addition to detection methods, Huang et al. [16] achieved grape cluster detection and instance segmentation in orchard environments based on an improved Mask R-CNN.

The above studies provide grape-cluster candidate regions and cluster-level spatial priors, but their outputs are mainly limited to the cluster level, making it difficult to directly perform BpGC counting or describe berry assignment relationships among adjacent clusters. For BpGC counting, cluster-level detection boxes or instance masks need to be further associated with berry-level localization results. Especially under grape cluster overlap, boundary adhesion, and local occlusion, relying solely on cluster-level region information makes it difficult to obtain stable intra-cluster counting results.

## 2.2. Grape Berry-Level Detection, Segmentation, and Counting

Grape berries are characterized by small size, large numbers, dense distributions, and frequent mutual occlusion. Therefore, berry-level studies usually adopt approaches such as instance segmentation, semantic segmentation, density-map regression, point-supervised counting, and semi-supervised learning [4–6,11,12,29]. Coviello et al. [34] transferred the idea of density estimation from crowd counting to field grape berry counting and proposed GBCNet for yield estimation. Zabawa et al. [35] used semantic segmentation and connected-component analysis to achieve grape berry detection and counting. Du and Liu [25] proposed AS-SwinT, which performs instance segmentation and counting of berries in individual grape clusters before thinning through Swin Transformer, feature fusion, and Soft-NMS. Li et al. [26] proposed CDMENet, introducing a density mutual-exclusion auxiliary task to alleviate dense occlusion and annotation-cost problems under semi-supervised conditions. Yang et al. [27] proposed Mask-GK, which uses mask Gaussian kernels to generate probability maps and combines them with watershed post-processing to achieve field grape berry segmentation and counting, while also releasing the GBISC dataset. Parr et al. [1] used smartphone RGB-D data for grape yield estimation, demonstrating the application potential of low-cost acquisition devices in berry analysis. Hao et al. [36] proposed WGMG-Net for postharvest grape quality assessment, preserving edge details under dense occlusion through discrete wavelet transform and simultaneously performing berry instance segmentation, counting, and compactness grading.

These methods improve berry-level visual analysis from different perspectives, but most studies mainly output berry counts for whole images, local regions, or single-cluster images. For multi-level grape counting, the berry-level branch must provide not only counts but also stable coordinates for intra-cluster assignment. Point-based counting frameworks such as P2PNet show that directly predicting object locations and counts based on point supervision can avoid intermediate errors introduced by density maps or pseudo-box construction, providing a useful reference for low-cost berry localization [32]. However, complex illumination, shadows, highlights, and occlusions in natural orchards can interfere with berry edges and local peak responses; therefore, point-based localization still requires stronger fine-grained structural representation.

## 2.3. BpGC Counting and Multi-Level Grape Counting

BpGC counting requires the simultaneous acquisition of grape-cluster regions, berry locations, and their correspondence, making it more complex than grape cluster detection or berry-level counting alone [4,11,12,30]. Yang et al. [9] proposed PMGDC, defining grape cluster detection, grape berry counting, and BpGC counting as three related subtasks. It predicts grape-cluster and berry probability maps using a neural network and applies multi-stage post-processing for joint inference. Du et al. [10] proposed MVDNet, which adopts a dual-branch network to perform berry density-map regression and grape cluster segmentation simultaneously and combines post-processing algorithms for intra-cluster

berry counting. Related studies indicate that multi-level grape counting requires the integration of cluster-level and berry-level information and the establishment of stable associations between them.

Existing multi-level methods still face two key problems. First, probability-map-based and density-map-based methods usually require post-processing to obtain object locations, cluster-level regions, or intra-cluster counts, and they are susceptible to response drift and regional boundary errors under occlusion, overlap, and boundary adhesion. Second, intra-cluster counting based on segmented regions usually assumes reliable geometric containment between berry points and grape-cluster regions. However, natural orchard images often contain locally undersized masks, boundary jitter, multiple overlapping masks, and berry points outside mask boundaries; therefore, simple in-box or in-mask rules can easily cause cross-cluster misassignment. General-purpose segmentation foundation models such as SAM provide a new technical basis for prompt-based instance mask generation [33], and their applications in grape cluster structure and berry trait analysis have also shown potential [3]. Therefore, stable assignment of berry point localization results to cluster-level instance masks remains a key problem in BpGC counting in natural orchards.

### 3. Materials and Methods

#### 3.1. Overall Framework

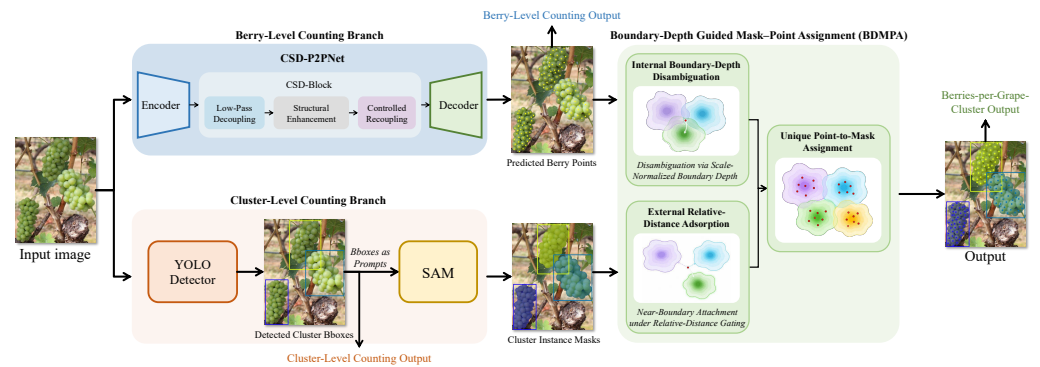
UniGC is designed for three interrelated subtasks with distinct outputs: berry-level counting, grape cluster-level detection and counting, and BpGC counting. As shown in Figure 1, given an input RGB image  $I \in \mathbb{R}^{H_I \times W_I \times 3}$ , the overall pipeline consists of three connected modules. First, the berry-level branch adopts CSD-P2PNet, a P2PNet-based point-supervised counting branch enhanced with the CSD-Block, to predict berry points for berry localization and whole-image berry counting. Second, the cluster-level branch uses a YOLO detector to predict grape-cluster detection boxes, which are then fed into SAM as box prompts to generate the corresponding grape-cluster instance masks. Finally, the BDMPA establishes assignments according to the spatial relationships between berry points and grape cluster instance masks, thereby outputting BpGC counting results.

The overall output is defined as:

$$\mathcal{O}(I) = \{\mathcal{P}, \hat{N}_b, \mathcal{B}, \mathcal{M}, \hat{N}_c, \hat{\mathbf{n}}\}. \quad (1)$$

where  $\mathcal{P} = \{p_i\}_{i=1}^N$  denotes the predicted berry point set, and  $p_i = (x_i, y_i)$  represents the image coordinates of the  $i$ -th berry point;  $\hat{N}_b$  denotes the whole-image berry-level counting result;  $\mathcal{B} = \{b_k\}_{k=1}^K$  denotes the set of grape cluster detection boxes;  $\mathcal{M} = \{M_k\}_{k=1}^K$  denotes the set of grape cluster instance masks, with  $M_k \subset \Omega$ , where  $\Omega$  represents the image pixel domain;  $\hat{N}_c$  denotes the grape cluster-level counting result; and  $\hat{\mathbf{n}} = (\hat{n}_1, \hat{n}_2, \dots, \hat{n}_K)$  represents the BpGC counting result vector.

With this design, UniGC decouples berry point prediction, grape-cluster instance region generation, and cross-level assignment, and performs berry-level counting, grape cluster-level detection and counting, and BpGC counting using berry point annotations and grape-cluster bounding-box annotations.

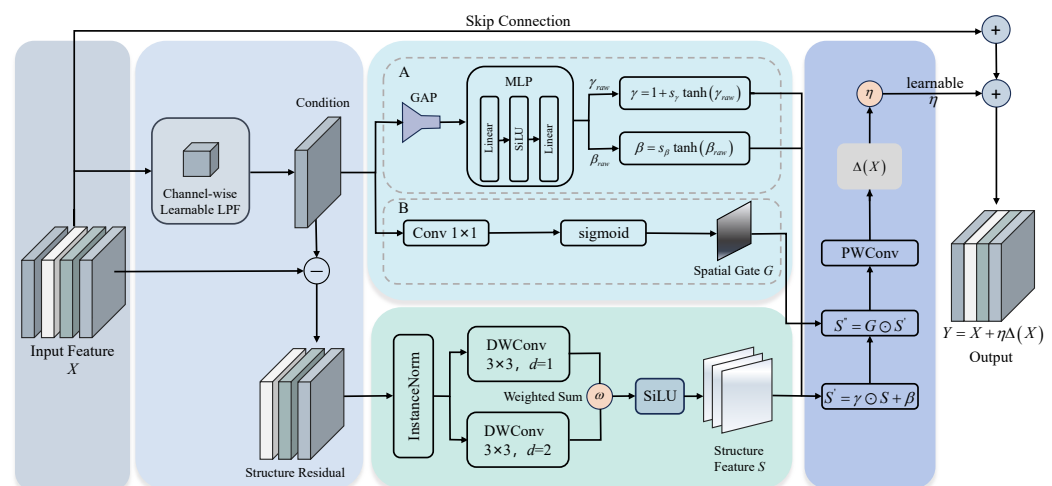


**Figure 1.** The proposed Unified Multi-Level Grape Counting (UniGC) framework.

### 3.2. CSD-P2PNet-Based Berry-Level Counting Branch

In natural orchard scenes, grape berry counting is highly sensitive to fine-grained structural cues such as local edges, texture changes, peak responses, and occlusion contours. However, disturbances such as strong illumination, shadows, uneven exposure, and highlight reflections can introduce slowly varying feature biases or local response shifts, which may interfere with the representation of berry-related geometric structures. If these condition-related variations are mixed with structure-sensitive responses in intermediate features, the model may become less stable in distinguishing true berry structures from background or illumination-induced disturbances.

To alleviate this problem, we introduce the CSD-Block into an intermediate feature layer of P2PNet, as shown in Figure 2. This module models intermediate features from a frequency-inspired perspective without applying an explicit Fourier transform. Instead, through learnable low-pass filtering and residual decomposition, it encourages the extraction of slowly varying condition-related components and residual structure-sensitive components in the spatial domain. The module then applies de-conditioned enhancement to the residual structural features and uses a condition-guided controlled recoupling mechanism to reintegrate structural information, thereby improving the robustness of berry point localization under complex illumination and occlusion.



**Figure 2.** Architecture of the Condition-Structure Decoupled Frequency Enhancement Block (CSD-Block).

#### 3.2.1. Explicit Decoupling of Condition and Structural Components

Let the input feature map be  $X \in \mathbb{R}^{B \times C \times H \times W}$ , where  $B$ ,  $C$ ,  $H$ , and  $W$  denote the batch size, number of channels, feature map height, and feature map width, respectively.

The output of CSD-Block has the same tensor shape as its input; therefore, the block can serve as a plug-and-play enhancement unit in the intermediate layers of P2PNet.

The module first uses a channel-wise learnable low-pass operator to extract a condition-related component as  $C_{\text{cond}} = \mathcal{L}(X)$ . Because the operator is constrained to perform spatial smoothing,  $C_{\text{cond}}$  is encouraged to capture slowly varying feature patterns that are associated with imaging conditions, such as illumination changes, exposure variations, and smooth background responses. The residual component is then obtained as  $S_0 = X - C_{\text{cond}}$ . Compared with the smoothed component,  $S_0$  is expected to retain more structure-sensitive responses related to berry localization, including local contrast changes, edge-like responses, texture variations, and occlusion contours.

The low-pass operator adopts depthwise convolution. Let the low-pass kernel size be  $k \times k$ . For each channel, the convolution kernel weights are obtained by applying softmax normalization to the kernel parameters  $\theta_c$ , namely  $W_c = \text{Softmax}(\theta_c)$  for  $c = 1, 2, \dots, C$ . Thus, any position  $(u, v)$  within the kernel satisfies  $W_c(u, v) \geq 0$  and  $\sum_{u,v} W_c(u, v) = 1$ .

This parameterization makes the convolution kernel naturally satisfy the constraints of non-negativity and unit-sum weights, thereby providing stable smoothing properties and helping extract slowly varying imaging-condition information from intermediate features.

### 3.2.2. De-Conditioned Structural Enhancement

Although  $S_0$  separates high-frequency structures to some extent, shadows and local uneven exposure may still affect the amplitude of structural responses. To reduce this condition-induced influence, CSD-Block applies Instance Normalization (IN) without affine parameters to  $S_0$ , yielding the de-conditioned structural feature  $\hat{S}$ . This operation can reduce mean drift and variance fluctuation caused by imaging-condition variations, allowing subsequent convolutions to focus more on the geometric structure itself.

On this basis, the module uses multi-dilation depthwise convolutions to extract structural details. Let the set of dilation rates be  $\mathcal{D} = \{d_1, d_2, \dots, d_M\}$ ; then the structural enhancement result is denoted as

$$S = \varphi \left( \sum_{m=1}^M \alpha_m \text{DWConv}_{3 \times 3}^{(d_m)}(\hat{S}) \right), \quad (2)$$

where  $\text{DWConv}_{3 \times 3}^{(d_m)}(\cdot)$  denotes the depthwise convolution (DWConv) with dilation rate  $d_m$ ,  $\varphi(\cdot)$  is a nonlinear activation function, and  $\alpha_m = \frac{\exp(\omega_m)}{\sum_{j=1}^M \exp(\omega_j)}$  is the branch fusion weight obtained by softmax normalization. In our implementation, we set  $\mathcal{D} = \{1, 2\}$ .

This structure can extract structural information such as local edges, texture changes, and occlusion contours from different receptive-field scales while maintaining relatively low computational complexity, thereby improving the stability of point responses in dense berry regions.

### 3.2.3. Condition-Guided Controlled Recoupling and Residual Write-Back

CSD-Block does not directly discard the condition component; instead, it allows imaging conditions to participate in structural enhancement in a limited and stable manner through a controlled recoupling mechanism. First, global average pooling (GAP) is applied to the condition component to obtain the condition descriptor vector  $z = \text{GAP}(C_{\text{cond}}) \in \mathbb{R}^{B \times C}$ . Subsequently, a two-layer multi-layer perceptron (MLP) predicts the channel-level scaling and bias terms  $[\gamma_{\text{raw}}, \beta_{\text{raw}}] = \text{MLP}(z)$ .

To limit the disturbance of condition modulation to the structural enhancement process, the hyperbolic tangent function and scale factors are used to constrain the modulation amplitude:

$$\gamma = 1 + s_\gamma \tanh(\gamma_{\text{raw}}), \quad \beta = s_\beta \tanh(\beta_{\text{raw}}), \quad (3)$$

where  $s_\gamma$  and  $s_\beta$  are modulation-amplitude coefficients. The channel-modulated structural feature is  $S' = \gamma \odot S + \beta$ , where  $\odot$  denotes element-wise multiplication, and  $\gamma$  and  $\beta$  are broadcast to the corresponding channels along the spatial dimensions.

Further, the module generates a single-channel spatial gating map from the condition component,  $G = \sigma(\text{Conv}_{1 \times 1}(C_{\text{cond}}))$ , where  $\sigma(\cdot)$  is the sigmoid function. The structural enhancement intensity is then constrained through spatial gating as  $S'' = G \odot S'$ .

This gating is intended to adaptively regulate the spatial strength of structural enhancement, especially in regions where condition-related variations such as highlights, shadow transitions, or local exposure changes may disturb berry-related responses. Finally, the module generates the enhancement residual using a  $1 \times 1$  pointwise convolution (PWConv),  $\Delta(X) = \text{PWConv}_{1 \times 1}(S'')$  and adds it to the original features in residual form as  $Y = X + \eta \Delta(X)$ , where  $\eta$  is a learnable scalar that controls the injection strength of the enhancement residual. Through the above process, CSD-Block selectively enhances structural information related to berry point localization while preserving the original feature representation.

### 3.2.4. Berry Point Prediction and Counting Output

The features enhanced by CSD-Block are then fed into the point-regression head of P2PNet to obtain the berry point set  $\mathcal{P} = \{p_i\}_{i=1}^N$ , with  $p_i = (x_i, y_i)$ . The berry-level count of the whole image is directly given by the number of predicted points,  $\hat{N}_b = |\mathcal{P}| = N$ .

This branch is trained using only berry point annotations and does not require berry-level pixel masks. Here,  $\mathcal{P}$  is used for subsequent point-mask assignment, and  $\hat{N}_b$  is used for whole-image berry-level counting evaluation.

## 3.3. Grape Cluster-Level Detection and Instance Mask Generation Branch

### 3.3.1. Grape Cluster Detection Box Prediction

The grape cluster-level branch is used to obtain candidate regions of grape clusters in an image and provide box prompts for instance mask generation. Given an input image  $I$ , the branch adopts a YOLO detector to predict the grape cluster detection-box set  $\mathcal{B} = g_{\theta_d}(I) = \{b_k\}_{k=1}^K$ , where  $b_k = (x_{1k}, y_{1k}, x_{2k}, y_{2k})$ ,  $g_{\theta_d}(\cdot)$  denotes the grape cluster detector, and  $K$  denotes the number of retained grape cluster instances.

### 3.3.2. Instance Mask Generation Under Detection-Box Prompts

Detection boxes alone cannot reliably characterize grape-cluster boundaries or provide sufficient geometric precision for subsequent point-to-cluster assignment. In particular, when the background inside a box is complex, adjacent grape clusters overlap, or boundary adhesion occurs, point-to-cluster assignment based on rectangular regions can easily lead to erroneous assignments. Therefore, UniGC feeds each grape cluster detection box  $b_k$  into SAM as a box prompt to generate the corresponding grape cluster instance mask  $M_k = h_{\theta_s}(I, b_k)$  for  $k = 1, 2, \dots, K$ , where  $h_{\theta_s}(\cdot)$  denotes the mask generator and  $M_k \subset \Omega$  represents the instance region of the  $k$ -th grape cluster in the image pixel domain. The corresponding binary indicator function is defined as:

$$\mathbf{1}_{M_k}(u, v) = \begin{cases} 1, & (u, v) \in M_k, \\ 0, & (u, v) \notin M_k, \end{cases} \quad (u, v) \in \Omega. \quad (4)$$

All grape cluster instance masks form the set  $\mathcal{M} = \{M_k\}_{k=1}^K$ .

In this process, YOLO provides grape cluster location candidates, and SAM then converts the detection-box prompts into instance-level regions, thereby providing finer geometric boundaries for subsequent BDMPA.

### 3.3.3. Grape Cluster Counting and Geometric Prior Output

After detection box prediction and instance mask generation, the grape cluster-level count is given by the number of retained instances,  $\hat{N}_c = |\mathcal{B}| = |\mathcal{M}| = K$ . The grape cluster-level branch finally outputs the grape cluster detection box set  $\mathcal{B}$ , the instance mask set  $\mathcal{M}$ , and the grape cluster-level counting result  $\hat{N}_c$ . Here,  $\mathcal{B}$  and  $\mathcal{M}$  serve as geometric priors for subsequent assignment modeling and are not used to directly perform intra-cluster berry counting.

## 3.4. Boundary-Depth Guided Mask-Point Assignment and BpGC Counting

### 3.4.1. Assignment Modeling Between Berry Points and Grape Cluster Masks

After obtaining the predicted berry point set  $\mathcal{P}$  and the grape-cluster instance mask set  $\mathcal{M}$ , UniGC uses BDMPA to assign each predicted berry point to at most one grape cluster instance. These assignments define a many-to-one point-to-cluster mapping. BDMPA follows the design principle of mandatory internal assignment and cautious external adsorption: for points located inside masks, unique disambiguation is preferentially completed among internal candidate instances; for points not covered by any mask, adsorption-based assignment is allowed only when they are located within the relative boundary neighborhood of an instance. This design alleviates point-to-cluster misassignment caused by adjacent grape-cluster overlap, boundary adhesion, locally undersized masks, and boundary jitter.

Let the predicted point set be  $\mathcal{P} = \{p_i\}_{i=1}^N$ , with  $p_i = (x_i, y_i)$ , and let the instance mask set be  $\mathcal{M} = \{M_k\}_{k=1}^K$ , with  $M_k \subset \Omega$ . Here,  $\Omega$  denotes the image pixel domain, and  $N$  and  $K$  denote the number of predicted berry points and the number of grape cluster instances, respectively. The objective of BDMPA is to output an assignment label  $a_i \in \{-1, 1, 2, \dots, K\}$  for each predicted point, where  $a_i = k$  indicates that point  $p_i$  is assigned to the  $k$ -th grape cluster instance, and  $a_i = -1$  indicates that the point is unassigned.

This assignment problem is difficult for two main reasons. On the one hand, adjacent grape clusters in natural orchard scenes often exhibit occlusion, adhesion, and local overlap, making it difficult to resolve assignment ambiguity in regions covered by multiple masks based solely on whether a point falls inside a mask. On the other hand, instance masks may exhibit boundary jitter, local holes, or undersized regions, causing some true berry points to fall outside all masks. Therefore, BDMPA performs point-to-cluster assignment using two complementary rules: internal boundary-depth disambiguation and external relative-distance adsorption.

To reduce the influence of image resolution and grape cluster scale differences on geometric assignment, a scale term is defined for the  $k$ -th mask as  $\sigma_k = \sqrt{|M_k|}$ , where  $|M_k|$  denotes the foreground pixel area of mask  $M_k$ . Let the minimum foreground bounding box of the  $k$ -th mask be  $B_k^M = (x_{1k}^M, y_{1k}^M, x_{2k}^M, y_{2k}^M)$ . The expanded boundary width is defined according to the relative adsorption threshold  $\rho$  as  $\delta_k = \lceil \rho \sigma_k + 2 \rceil$ , and the expanded local region of interest (ROI) is obtained as  $R_k = \text{expand}(B_k^M, \delta_k)$ .

To reduce computational overhead, BDMPA does not compute a global distance field for each mask over the entire image; instead, it computes the Euclidean distance transform within the local ROI corresponding to each mask. To avoid truncation effects on the internal boundary depth caused by the cropped boundary of the local ROI, a one-pixel ring of background pixels is padded outside the local binary mask before computing the distance field.

Let the internal boundary-depth field and the external distance field be denoted by  $D_k^{\text{in}}$  and  $D_k^{\text{out}}$ , respectively. For any pixel position  $x$ , they are defined as

$$D_k^{\text{in}}(x) = \begin{cases} d(x, \partial M_k), & x \in M_k, \\ 0, & x \notin M_k, \end{cases} \quad (5)$$

$$D_k^{\text{out}}(x) = \begin{cases} 0, & x \in M_k, \\ d(x, M_k), & x \notin M_k, \end{cases} \quad (6)$$

where  $d(\cdot, \cdot)$  denotes the Euclidean distance, and  $\partial M_k$  denotes the mask boundary. Scale-normalized distances are further constructed as

$$\tilde{D}_k^{\text{in}}(x) = \frac{D_k^{\text{in}}(x)}{\sigma_k + \varepsilon}, \quad \tilde{D}_k^{\text{out}}(x) = \frac{D_k^{\text{out}}(x)}{\sigma_k + \varepsilon}, \quad (7)$$

where  $\varepsilon$  is a numerical stability term. The above scale normalization makes the internal boundary depth and external distance of grape cluster instances with different sizes comparable in a relative scale space.

### 3.4.2. Boundary-Depth Disambiguation for Internal Points

For a predicted point  $p_i = (x_i, y_i)$ , its continuous coordinates are first mapped to the nearest discrete pixel position as  $\hat{x}_i = \text{round}(x_i)$  and  $\hat{y}_i = \text{round}(y_i)$ ; thus,  $\hat{p}_i = (\hat{x}_i, \hat{y}_i)$ . If  $\hat{p}_i$  lies outside the image boundary,  $a_i = -1$  is directly assigned. For valid points, the internal candidate set is constructed as  $\mathcal{C}_i^{\text{in}} = \{k \mid \hat{p}_i \in M_k\}$ .

If  $\mathcal{C}_i^{\text{in}} \neq \emptyset$ , the point is located inside one or more masks. In this case, BDMPPA adopts the boundary-depth maximization principle and assigns the point to the mask with the largest normalized internal boundary depth:

$$a_i = \arg \max_{k \in \mathcal{C}_i^{\text{in}}} \tilde{D}_k^{\text{in}}(\hat{p}_i). \quad (8)$$

The rationale for this rule is as follows: if a point is located inside a mask and closer to its central region, its distance to the mask boundary is usually larger; if a point falls within another mask only because of a small overlap caused by adjacent-instance overlap or boundary errors, its internal boundary depth is usually smaller. Therefore, the normalized internal boundary depth can serve as a geometrically interpretable basis for unique assignment in overlapping regions. Once a point is assigned within the internal candidate set, it does not enter the external association stage, thereby avoiding duplicate assignment.

### 3.4.3. Relative-Distance Adsorption for External Points

If  $\mathcal{C}_i^{\text{in}} = \emptyset$ , point  $p_i$  is not covered by any grape cluster mask. Directly discarding such points may lead to undercounting of berries near mask edges; therefore, BDMPPA further performs external relative-distance adsorption. This process only considers masks whose expanded ROI covers the point and whose normalized external distance does not exceed the threshold. The external candidate set is defined as

$$\mathcal{C}_i^{\text{out}} = \left\{ k \mid \hat{p}_i \in R_k, \tilde{D}_k^{\text{out}}(\hat{p}_i) \leq \rho \right\}. \quad (9)$$

If  $\mathcal{C}_i^{\text{out}} \neq \emptyset$ , the point is assigned to the mask with the smallest normalized external distance:

$$a_i = \arg \min_{k \in \mathcal{C}_i^{\text{out}}} \tilde{D}_k^{\text{out}}(\hat{p}_i). \quad (10)$$

Otherwise,  $a_i = -1$ , indicating that the point is not assigned to any grape-cluster instance. The parameter  $\rho$  controls the allowable relative distance range for adsorption: a smaller  $\rho$  helps reduce long-distance erroneous adsorption but may retain more unassigned points, whereas a larger  $\rho$  can adsorb more points outside boundaries but increases the risk of incorrect assignment. Since the external distance has been scale-normalized by  $\sigma_k$ ,  $\rho$  represents the adsorption range relative to the instance scale rather than a fixed pixel distance.

#### 3.4.4. Unified Decision Rule and BpGC Output

Considering both internal disambiguation and external adsorption, the final decision rule of BDMPA can be written as

$$a_i = \begin{cases} \arg \max_{k \in \mathcal{C}_i^{\text{in}}} \tilde{D}_k^{\text{in}}(\hat{p}_i), & \mathcal{C}_i^{\text{in}} \neq \emptyset, \\ \arg \min_{k \in \mathcal{C}_i^{\text{out}}} \tilde{D}_k^{\text{out}}(\hat{p}_i), & \mathcal{C}_i^{\text{in}} = \emptyset, \mathcal{C}_i^{\text{out}} \neq \emptyset, \\ -1, & \text{otherwise.} \end{cases} \quad (11)$$

After point-mask assignment is completed, the berry count of the  $k$ -th grape cluster is defined as  $\hat{n}_k = \sum_{i=1}^N \mathbf{1}(a_i = k)$  for  $k = 1, 2, \dots, K$ , where  $\mathbf{1}(\cdot)$  is the indicator function. Finally, the BpGC counting result vector is  $\hat{\mathbf{n}} = (\hat{n}_1, \hat{n}_2, \dots, \hat{n}_K)$ .

Through this process, BDMPA integrates the point-level prediction results  $\mathcal{P}$  and cluster-level instance masks  $\mathcal{M}$  within the same geometric assignment space, completing BpGC counting without introducing additional trainable parameters.

#### 3.5. Dataset and Annotation Organization

The experiments in this study use a mixed grape dataset constructed from WGSD and GBISC. WGSD was collected from the vineyard of Guaspari Winery in São Paulo State, Brazil, with images acquired in April 2017 and April 2018, and covers five wine grape cultivars: Chardonnay, Cabernet Franc, Cabernet Sauvignon, Sauvignon Blanc, and Syrah [13]. GBISC was collected in July 2023 from vineyards in Longquanyi District and Shuangliu District, Chengdu, Sichuan Province, China, and covers three grape cultivars: Kyoho, Shine Muscat, and Summer Black [27]. After annotation unification and validity filtering, the organized dataset contains 450 RGB images, 6152 grape-cluster bounding boxes, and 237,696 berry center points.

The dataset was randomly divided at the image level into 360 training images and 90 validation images. The training split contains 240 WGSD images and 120 GBISC images, while the validation split contains 60 WGSD images and 30 GBISC images. For model training, the organized dataset provides berry center-point annotations for berry-level counting and grape-cluster bounding-box annotations for cluster-level detection.

For BpGC evaluation, point-to-cluster correspondence labels were additionally annotated for the validation split, indicating which grape cluster each annotated berry point belongs to. These correspondence labels cover 1258 ground-truth grape clusters and 46,292 berry center points in the validation split and are used only for BpGC evaluation. The detailed statistics of the training and validation splits are summarized in Table 1.

**Table 1.** Statistics of the final organized grape dataset used in the experiments.

| Split | Source | Images | Cluster Boxes | Berry Center Points |
|-------|--------|--------|---------------|---------------------|
| Train | WGSD   | 240    | 3563          | 150,469             |
|       | GBISC  | 120    | 1331          | 40,935              |

**Table 1.** *Cont.*

| Split | Source | Images | Cluster Boxes | Berry Center Points |
|-------|--------|--------|---------------|---------------------|
| Val   | WGISD  | 60     | 951           | 36,837              |
|       | GBISC  | 30     | 307           | 9455                |
| Total | –      | 450    | 6152          | 237,696             |

To quantify the annotation effort associated with the supervision formats used in this study, three annotators participated in a timed annotation study. Each annotator labeled 15 randomly selected WGISD images and 15 GBISC images, resulting in 90 annotated images in total. The mean operation times were approximately 0.8 s per berry center point, 1.4 s per grape-cluster bounding box, 2.6 s per berry instance mask, and 10.6 s per grape-cluster instance mask. Based on the numbers of berries and clusters in the complete 450-image dataset, the average annotation time was estimated to be 7.36 min per image for berry center points and grape-cluster bounding boxes, compared with 25.30 min per image for dual instance masks. Thus, dual instance-mask annotation required approximately 3.44 times as much time as point-and-box annotation. These estimates quantify annotation time rather than monetary cost, and the point-to-cluster correspondence labels used for BpGC evaluation were not included in the training supervision. In addition to the measured time difference, UniGC requires simpler supervision formats: its trainable branches rely only on berry center points and grape-cluster bounding boxes, without pixel-level instance masks. In regions with overlapping clusters, touching berries, or occlusion, initial masks generated using interactive segmentation tools generally require repeated inspection, boundary correction, separation of touching instances, and removal of mask overlaps. In comparison, center points and bounding boxes are more direct to create, inspect, and revise, facilitating the organization of large-scale grape image datasets.

## 4. Experiments and Results

### 4.1. Experimental Setup

All experiments in this study were conducted on the same server. The hardware platform was configured with an Intel(R) Xeon(R) Platinum 8481C CPU and an NVIDIA GeForce RTX 4090D GPU with 24 GB of memory. During UniGC training, the cluster-level detection branch used YOLO12-S [37] by default and was trained for 200 epochs with a batch size of 8, an input image size of  $1024 \times 1024$  pixels, SGD as the optimizer, an initial learning rate of 0.01, a momentum of 0.937, and a weight decay of 0.0005. The SAM-based mask generation module was used only as a prompt-driven inference module: the detection boxes predicted by YOLO are fed into the pretrained SAM to generate grape-cluster instance masks, and no SAM training or fine-tuning is performed. The berry-level counting branch uses the proposed CSD-P2PNet, which follows the training protocol of P2PNet [32]. Specifically, CSD-Block is inserted into the intermediate feature layer of P2PNet, while the original point-regression training objective and prediction head are retained. This branch was trained for 500 epochs, with a batch size of 4, Adam as the optimizer, and a learning rate of  $1.0 \times 10^{-4}$ . To improve training robustness, YOLO12-S was trained using the official settings and data augmentation strategies, including HSV perturbation, translation, scaling, horizontal flipping, Mosaic, RandAugment, and random erasing. CSD-P2PNet followed the data augmentation settings of P2PNet, including random scaling, random cropping, and random horizontal flipping. During inference and evaluation, the YOLO confidence threshold is set to 0.25, and the relative adsorption threshold of BDMPA is set to  $\rho = 0.03$ ; in BpGC evaluation, intersection over union (IoU) is set to 0.6 as the matching threshold between ground-truth (GT) grape clusters and predicted grape clusters. To assess

the robustness of the reported results to stochastic variation, selected key experiments were repeated using three distinct random seeds, while all remaining experiments were conducted using a fixed random seed.

## 4.2. Evaluation Metrics

### 4.2.1. Grape Berry Counting

To evaluate whole-image grape berry counting performance, this study uses mean absolute error (MAE) and root mean square error (RMSE). They are calculated as follows:

$$\text{MAE} = \frac{1}{N_{\text{img}}} \sum_{i=1}^{N_{\text{img}}} |C_i - C_i^{\text{GT}}|, \quad (12)$$

$$\text{RMSE} = \sqrt{\frac{1}{N_{\text{img}}} \sum_{i=1}^{N_{\text{img}}} (C_i - C_i^{\text{GT}})^2}, \quad (13)$$

where  $N_{\text{img}}$  denotes the number of evaluation images, and  $C_i$  and  $C_i^{\text{GT}}$  denote the predicted and ground-truth berry counts in the  $i$ -th image, respectively. Lower MAE and RMSE indicate smaller whole-image berry-level counting errors.

### 4.2.2. Grape Cluster Detection and Berry Point Localization

To evaluate grape cluster detection performance, this study uses precision, recall,  $F_1$ , mAP@0.5, and mAP@0.5:0.95. Berry point localization is additionally evaluated using precision, recall,  $F_1$ , mean pixel localization error, and normalized localization error. Precision measures the accuracy of predicted boxes, recall measures the completeness with which ground-truth grape clusters are detected, and  $F_1$  summarizes the balance between them:

$$F_1 = \frac{2 \times P \times R}{P + R}, \quad (14)$$

where  $P$  and  $R$  denote precision and recall, respectively. mAP@0.5 denotes the mean average precision at IoU = 0.5, whereas mAP@0.5:0.95 denotes the mean average precision over IoU thresholds from 0.50 to 0.95 with a step size of 0.05. Higher values of the above detection metrics indicate better grape cluster detection performance.

For berry point localization, let  $\mathcal{P} = \{p_i\}_{i=1}^{N_p}$  and  $\mathcal{G} = \{g_j\}_{j=1}^{N_g}$  denote the predicted and ground-truth berry centers in an image, respectively. The Euclidean distance between predicted point  $p_i$  and ground-truth point  $g_j$  is defined as

$$d_{ij} = \|p_i - g_j\|_2. \quad (15)$$

To account for spatial-density variation, each ground-truth point  $g_j$  is assigned a local scale  $s_j$ , defined as the mean distance from  $g_j$  to its  $k_j = \min(3, N_g - 1)$  nearest other ground-truth points. The matching cost is then normalized as

$$\tilde{d}_{ij} = \frac{d_{ij}}{s_j + 10^{-12}}. \quad (16)$$

A global one-to-one assignment between predicted and ground-truth points is obtained using the Hungarian algorithm by minimizing the total normalized matching cost. An assigned pair is accepted as a successful match when  $\tilde{d}_{ij} \leq 0.5$ . Accepted pairs are counted as true positives (TP), whereas unmatched predicted points and unmatched ground-truth points are counted as false positives (FP) and false negatives (FN), respec-

tively. Point-level precision, recall, and  $F_1$  are then calculated using the same definitions given above.

Let  $\mathcal{A}$  denote the set of accepted point pairs. The mean pixel localization error and normalized localization error are calculated only over the accepted pairs as

$$E_{\text{pixel}} = \frac{1}{|\mathcal{A}|} \sum_{(i,j) \in \mathcal{A}} d_{ij}, \quad E_{\text{norm}} = \frac{1}{|\mathcal{A}|} \sum_{(i,j) \in \mathcal{A}} \tilde{d}_{ij}. \quad (17)$$

At the dataset level, TP, FP, and FN are accumulated over all evaluation images before calculating precision, recall, and  $F_1$ , whereas  $E_{\text{pixel}}$  and  $E_{\text{norm}}$  are averaged over all accepted point pairs. Higher precision, recall, and  $F_1$  values, together with lower localization errors, indicate better berry point localization performance. Subsequent BpGC evaluation used IoU = 0.6 as the ground-truth-prediction matching threshold.

#### 4.2.3. BpGC Counting

BpGC counting is evaluated from both matched-cluster and all-cluster perspectives. Following the matched-cluster protocol used in PMGDC [9], a GT grape cluster is considered successfully matched if there exists at least one predicted grape cluster whose intersection over union (IoU) with it exceeds 0.6. When multiple predicted grape clusters satisfy this condition for the same GT grape cluster, the predicted cluster with the largest IoU is used to provide the corresponding predicted BpGC count. Let the number of matched GT grape clusters be  $N_{\text{match}}$ , and let the total number of GT grape clusters be  $N_{\text{GT}}$ . The matching rate is defined as  $R_{\text{match}} = \frac{N_{\text{match}}}{N_{\text{GT}}}$ . For the  $j$ -th matched GT grape cluster, let  $g_j$  and  $\hat{g}_j$  denote the ground-truth and predicted berry counts, respectively. Following the parts-per-object counting metrics used in agricultural phenotyping studies [38], this study uses mean relative deviation (MRD) and one minus the fraction of variance unexplained ( $1 - \text{FVU}$ ) to evaluate BpGC counting performance on the matched grape clusters, characterizing relative counting deviation and the degree to which true intra-cluster count variation is explained, respectively:

$$\text{MRD} = \frac{1}{N_{\text{match}}} \sum_{j=1}^{N_{\text{match}}} \frac{|\hat{g}_j - g_j|}{g_j}, \quad (18)$$

$$1 - \text{FVU} = 1 - \frac{\sum_{j=1}^{N_{\text{match}}} (g_j - \hat{g}_j)^2}{\sum_{j=1}^{N_{\text{match}}} (g_j - \bar{g})^2}, \quad (19)$$

where  $\bar{g}$  denotes the mean ground-truth berry count of the matched grape clusters. A higher  $R_{\text{match}}$  indicates greater grape-cluster matching coverage. A lower MRD indicates a smaller relative counting deviation on matched grape clusters, whereas a higher  $1 - \text{FVU}$  indicates that the predicted BpGC counts explain a larger fraction of the true intra-cluster count variation.

Because the matched-cluster metrics exclude ground-truth clusters that are not associated with any prediction, we additionally introduce the all-cluster penalized mean relative deviation (ACP-MRD) to evaluate overall BpGC performance across matched and unmatched clusters. Let  $N_{\text{pred}}$  denote the total number of predicted grape clusters after post-processing, and let  $N_{\text{used}}$  denote the number of unique predictions appearing in the matched pairs. The numbers of missed ground-truth clusters, unused predictions, and prediction reuse events are respectively defined as

$$N_{\text{miss}} = N_{\text{GT}} - N_{\text{match}}, \quad (20)$$

$$N_{\text{extra}} = N_{\text{pred}} - N_{\text{used}}, \quad (21)$$

and

$$N_{\text{reuse}} = N_{\text{match}} - N_{\text{used}}. \quad (22)$$

The accumulated relative counting error over the matched pairs is

$$E_{\text{match}} = \sum_{j=1}^{N_{\text{match}}} \frac{|\hat{g}_j - g_j|}{g_j} = N_{\text{match}} \text{MRD}. \quad (23)$$

ACP-MRD is then defined as

$$\text{ACP-MRD} = \frac{E_{\text{match}} + N_{\text{miss}} + N_{\text{extra}} + N_{\text{reuse}}}{N_{\text{GT}}}. \quad (24)$$

A lower ACP-MRD indicates better overall BpGC performance after jointly accounting for matched-cluster counting errors, missed ground-truth clusters, unused predictions, and prediction reuse.

#### 4.2.4. Grape-Cluster Mask Quality

To directly assess the quality of grape-cluster masks generated from box prompts, we use mean intersection over union (mIoU), boundary  $F_1$ , area log error, and mask failure rate. For  $N$  evaluated mask pairs, let  $M_i$  and  $G_i$  denote the predicted and ground-truth masks of the  $i$ -th instance, respectively. The instance-level IoU and mIoU are defined as

$$\text{IoU}_i = \frac{|M_i \cap G_i|}{|M_i \cup G_i|}, \quad \text{mIoU} = \frac{1}{N} \sum_{i=1}^N \text{IoU}_i. \quad (25)$$

Boundary  $F_1$  is the harmonic mean of boundary precision and boundary recall, calculated using the same boundary-matching tolerance for all evaluated models:

$$F_1^{\text{boundary}} = \frac{2P_{\text{boundary}}R_{\text{boundary}}}{P_{\text{boundary}} + R_{\text{boundary}}}. \quad (26)$$

The area log error and mask failure rate are defined as

$$E_{\text{area}} = \frac{1}{N} \sum_{i=1}^N \left| \log \frac{|M_i| + \epsilon}{|G_i| + \epsilon} \right|, \quad R_{\text{fail}} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\text{IoU}_i < 0.5), \quad (27)$$

where  $\epsilon$  is a small constant used for numerical stability and  $\mathbb{I}(\cdot)$  is the indicator function. The area log error measures the multiplicative discrepancy between predicted and ground-truth mask areas, whereas the failure rate quantifies the proportion of masks with an IoU below 0.5. Higher mIoU and boundary  $F_1$ , together with lower area log error and failure rate, indicate better mask quality.

Mask quality is evaluated under two prompt settings: ground-truth grape-cluster boxes and fixed YOLO12-S predicted boxes matched to ground-truth clusters. The ground-truth-box setting isolates the mask generator from detector localization errors, whereas the predicted-box setting reflects mask quality under the box prompts used in the UniGC pipeline.

#### 4.3. Comparison with Existing Methods

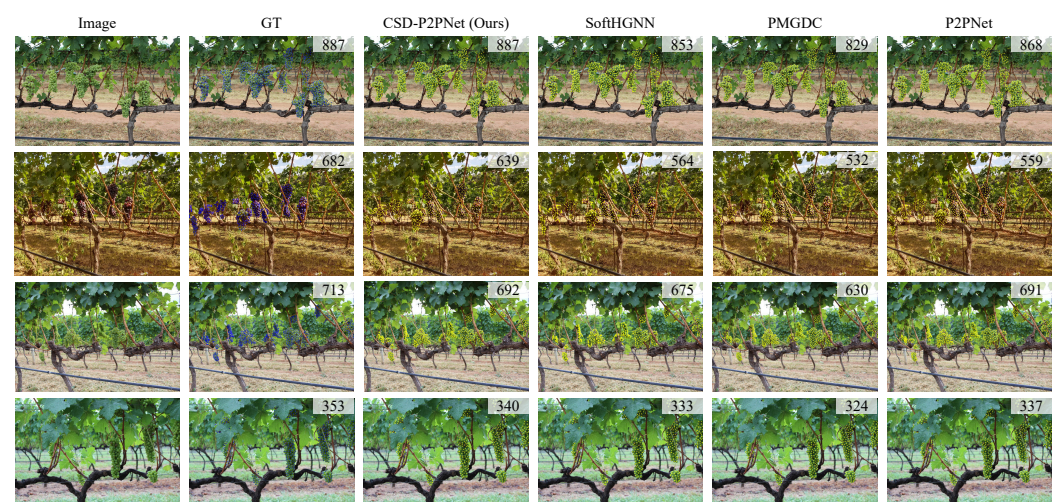
This section compares UniGC with representative existing methods on the mixed dataset from two perspectives: (i) whole-image berry counting and berry point localization, and (ii) multi-level grape detection and BpGC counting.

#### 4.3.1. Comparison of Whole-Image Berry Counting and Berry Point Localization

For whole-image berry counting, CSD-P2PNet achieved the lowest MAE and RMSE among the compared methods, with values of 23.3 and 33.2, respectively. Table 2 presents the quantitative comparison, and Figure 3 shows representative predictions. Relative to the second-best result for each metric, CSD-P2PNet reduced MAE by 14.0% compared with APGCC [39] and reduced RMSE by 11.9% compared with P2PNet [32].

**Table 2.** Comparison of whole-image berry-level counting results.

| Year | Method            | MAE  | RMSE |
|------|-------------------|------|------|
| 2021 | P2PNet [32]       | 27.7 | 37.7 |
| 2021 | GL [40]           | 38.6 | 53.5 |
| 2022 | CLTR [41]         | 32.5 | 41.6 |
| 2023 | PET-main [42]     | 34.0 | 59.9 |
| 2024 | PMGDC [9]         | 35.1 | 48.5 |
| 2024 | APGCC [39]        | 27.1 | 38.9 |
| 2024 | WSSC_TAF [43]     | 31.6 | 49.5 |
| 2025 | DSGC-Net [44]     | 31.3 | 40.4 |
| 2026 | SoftHGNN [45]     | 27.5 | 38.3 |
| 2026 | CSD-P2PNet (Ours) | 23.3 | 33.2 |



**Figure 3.** Visualization of grape berry-level counting results using different methods.

To further assess robustness and point-localization quality, PMGDC [9], P2PNet [32], and CSD-P2PNet were evaluated over three runs using different random seeds. Table 3 reports the results as mean  $\pm$  standard deviation (SD).

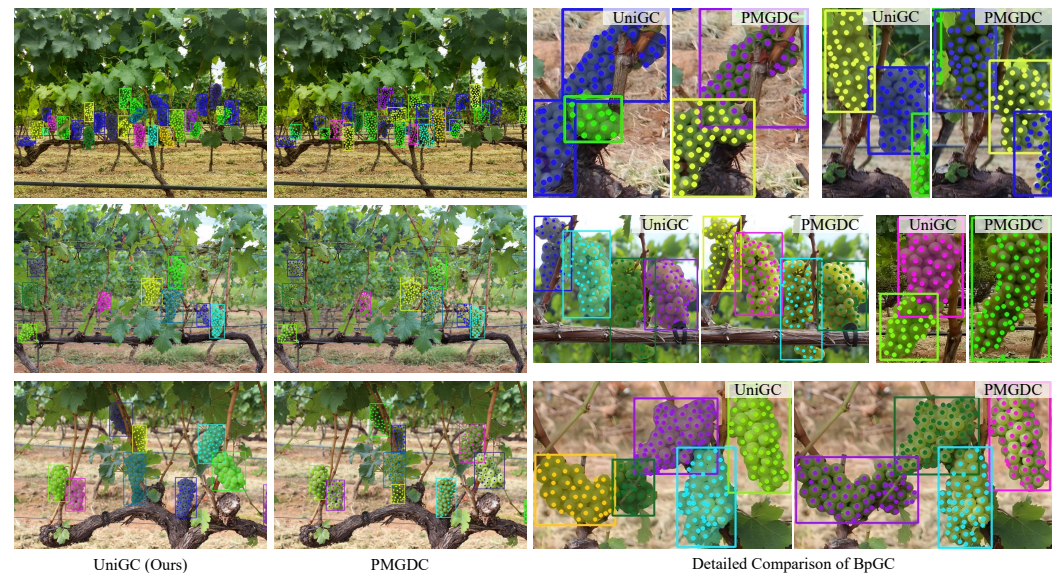
Across the three runs, CSD-P2PNet achieved an MAE of  $23.4407 \pm 0.1577$  and an RMSE of  $32.8898 \pm 1.2682$ . Relative to PMGDC [9] and P2PNet [32], it reduced MAE by 36.58% and 14.33%, and RMSE by 32.54% and 9.15%, respectively. For point localization, CSD-P2PNet achieved the highest mean precision and  $F_1$  score, together with the lowest pixel and normalized localization errors. PMGDC [9] achieved the highest mean recall, while CSD-P2PNet remained close to PMGDC [9] and exceeded P2PNet [32]. Relative to PMGDC [9] and P2PNet [32], CSD-P2PNet reduced the pixel localization error by 29.80% and 25.26%, and the normalized localization error by 30.28% and 27.96%, respectively.

#### 4.3.2. Comparison of Multi-Level Detection and Counting

UniGC was compared with PMGDC [9] for grape-cluster detection and BpGC counting. Table 4 reports the representative single-run results. UniGC achieved a precision of 0.879, a recall of 0.809, and an  $F_1$  score of 0.843. Relative to PMGDC [9], UniGC increased

mAP@0.5 from 0.747 to 0.880 and mAP@0.5:0.95 from 0.402 to 0.656, corresponding to relative improvements of 17.8% and 63.2%, respectively. These gains provide more reliable grape-cluster locations for subsequent mask generation and point-to-cluster assignment.

For BpGC counting, UniGC matched 996 grape clusters, compared with 989 for PMGDC [9]. It reduced the matched-cluster MRD from 0.1584 to 0.1336 and increased  $1 - \text{FVU}$  from 0.8193 to 0.8557, corresponding to relative improvements of 15.7% and 4.4%, respectively. Figure 4 presents the corresponding qualitative comparison.



**Figure 4.** Visualization of multi-level detection and BpGC results using PMGDC and our proposed UniGC.

To assess robustness to stochastic variation, PMGDC [9] and UniGC were further evaluated over three runs using different random seeds. Table 5 reports the results as mean  $\pm$  standard deviation (SD).

Across the three runs, UniGC achieved a precision of  $0.8948 \pm 0.0157$ , a recall of  $0.7975 \pm 0.0107$ , and an  $F_1$  score of  $0.8433 \pm 0.0056$ . Relative to PMGDC [9], these metrics increased by 9.50%, 1.64%, and 5.35%, respectively. UniGC also improved mAP@0.5 and mAP@0.5:0.95 by 18.34% and 69.27%, respectively. For BpGC counting, relative to PMGDC [9], UniGC reduced the matched-cluster MRD by 20.15%, increased  $1 - \text{FVU}$  by 3.91%, and reduced ACP-MRD by 2.43%. The repeated-run results therefore support the robustness of the improvements in grape-cluster detection, matched-cluster counting, and all-cluster penalized evaluation.

#### 4.3.3. Source-Stratified Evaluation on WGISD and GBISC

To examine whether the comparative trends were consistent across the two data sources, the 90-image evaluation set was stratified into the WGISD and GBISC subsets. PMGDC [9] and UniGC were evaluated over three runs using the same random-seed settings, and the results are reported as mean  $\pm$  standard deviation in Table 6. Because both subsets are drawn from the existing evaluation set, this analysis characterizes source-specific performance variation rather than independent cross-dataset generalization.

**Table 3.** Three-run comparison of representative methods for berry counting and point localization.

| Method            | Whole-Image Counting |                      | Point Localization  |                     |                     |                     |                     |
|-------------------|----------------------|----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                   | MAE                  | RMSE                 | Prec.               | Rec.                | $F_1$               | Pix. Err.           | Norm. Err.          |
| PMGDC [9]         | 36.9630 $\pm$ 1.8003 | 48.7579 $\pm$ 0.3782 | 0.8315 $\pm$ 0.0080 | 0.8419 $\pm$ 0.0294 | 0.8366 $\pm$ 0.0184 | 5.5102 $\pm$ 0.2859 | 0.1866 $\pm$ 0.0096 |
| P2PNet [32]       | 27.3630 $\pm$ 0.9470 | 36.2017 $\pm$ 1.4463 | 0.8111 $\pm$ 0.0222 | 0.8150 $\pm$ 0.0137 | 0.8130 $\pm$ 0.0175 | 5.1758 $\pm$ 0.0519 | 0.1806 $\pm$ 0.0039 |
| CSD-P2PNet (Ours) | 23.4407 $\pm$ 0.1577 | 32.8898 $\pm$ 1.2682 | 0.8370 $\pm$ 0.0095 | 0.8378 $\pm$ 0.0107 | 0.8374 $\pm$ 0.0097 | 3.8682 $\pm$ 0.0432 | 0.1301 $\pm$ 0.0012 |

**Table 4.** Comparison of multi-level detection and BpGC counting performance between UniGC and PMGDC.

| Method       | Prec. | Rec.  | $F_1$ | mAP@0.5 | mAP@0.5:0.95 | $N_{\text{match}}$ | $R_{\text{match}}$ | MRD    | 1 – FVU |
|--------------|-------|-------|-------|---------|--------------|--------------------|--------------------|--------|---------|
| PMGDC [9]    | 0.817 | 0.785 | 0.801 | 0.747   | 0.402        | 989                | 0.786              | 0.1584 | 0.8193  |
| UniGC (Ours) | 0.879 | 0.809 | 0.843 | 0.880   | 0.656        | 996                | 0.792              | 0.1336 | 0.8557  |

**Table 5.** Three-run comparison of PMGDC and UniGC for grape-cluster detection and BpGC counting.

| Method       | Grape-Cluster Detection |                 |                 |                 |                 | BpGC Counting   |                 |                 |
|--------------|-------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
|              | Prec.                   | Rec.            | $F_1$           | mAP@0.5         | mAP@0.5:0.95    | MRD             | 1 – FVU         | ACP-MRD         |
| PMGDC [9]    | 0.8172 ± 0.0115         | 0.7846 ± 0.0036 | 0.8005 ± 0.0037 | 0.7427 ± 0.0051 | 0.3889 ± 0.0130 | 0.1613 ± 0.0034 | 0.8266 ± 0.0089 | 0.5188 ± 0.0105 |
| UniGC (Ours) | 0.8948 ± 0.0157         | 0.7975 ± 0.0107 | 0.8433 ± 0.0056 | 0.8789 ± 0.0080 | 0.6583 ± 0.0024 | 0.1288 ± 0.0043 | 0.8589 ± 0.0098 | 0.5062 ± 0.0124 |

**Table 6.** Source-stratified three-run comparison of PMGDC and UniGC on the WGISD and GBISC subsets of the 90-image evaluation set.

| Panel A. Whole-Image Berry Counting and Point Localization |              |        |                       |                       |                     |                     |                     |                     |                     |                     |
|------------------------------------------------------------|--------------|--------|-----------------------|-----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Source                                                     | Method       | Images | MAE                   | RMSE                  | Prec.               | Rec.                | $F_1$               | Pix. Err.           | Norm. Err.          |                     |
| WGISD                                                      | PMGDC [9]    | 60     | 40.9111 $\pm$ 4.7567  | 52.9941 $\pm$ 5.6892  | 0.8142 $\pm$ 0.0102 | 0.8454 $\pm$ 0.0282 | 0.8294 $\pm$ 0.0189 | 4.6415 $\pm$ 0.3027 | 0.2021 $\pm$ 0.0117 |                     |
| WGISD                                                      | UniGC (Ours) | 60     | 28.6833 $\pm$ 1.7765  | 38.5428 $\pm$ 3.1496  | 0.8352 $\pm$ 0.0055 | 0.8392 $\pm$ 0.0094 | 0.8372 $\pm$ 0.0071 | 3.2461 $\pm$ 0.0260 | 0.1403 $\pm$ 0.0012 |                     |
| GBISC                                                      | PMGDC [9]    | 30     | 29.0667 $\pm$ 14.5909 | 35.9964 $\pm$ 16.2773 | 0.8439 $\pm$ 0.0270 | 0.8322 $\pm$ 0.0223 | 0.8380 $\pm$ 0.0245 | 8.9283 $\pm$ 0.3820 | 0.1259 $\pm$ 0.0049 |                     |
| GBISC                                                      | UniGC (Ours) | 30     | 11.9667 $\pm$ 1.5762  | 15.8404 $\pm$ 1.8730  | 0.9087 $\pm$ 0.0095 | 0.8282 $\pm$ 0.0352 | 0.8662 $\pm$ 0.0161 | 6.3441 $\pm$ 0.2400 | 0.0895 $\pm$ 0.0020 |                     |
| Panel B. Grape-Cluster Detection and BpGC Counting         |              |        |                       |                       |                     |                     |                     |                     |                     |                     |
| Source                                                     | Method       | Images | Prec.                 | Rec.                  | $F_1$               | AP50                | AP50:95             | MRD                 | 1 – FVU             | ACP-MRD             |
| WGISD                                                      | PMGDC [9]    | 60     | 0.8173 $\pm$ 0.0087   | 0.7834 $\pm$ 0.0074   | 0.7999 $\pm$ 0.0048 | 0.7449 $\pm$ 0.0117 | 0.4161 $\pm$ 0.0051 | 0.1716 $\pm$ 0.0041 | 0.8143 $\pm$ 0.0132 | 0.5497 $\pm$ 0.0126 |
| WGISD                                                      | UniGC (Ours) | 60     | 0.8718 $\pm$ 0.0148   | 0.7788 $\pm$ 0.0209   | 0.8224 $\pm$ 0.0052 | 0.8502 $\pm$ 0.0090 | 0.6190 $\pm$ 0.0050 | 0.1245 $\pm$ 0.0025 | 0.8509 $\pm$ 0.0208 | 0.5265 $\pm$ 0.0129 |
| GBISC                                                      | PMGDC [9]    | 30     | 0.8149 $\pm$ 0.0305   | 0.7861 $\pm$ 0.0167   | 0.8001 $\pm$ 0.0218 | 0.7619 $\pm$ 0.0209 | 0.3667 $\pm$ 0.0334 | 0.1438 $\pm$ 0.0089 | 0.8411 $\pm$ 0.0180 | 0.4947 $\pm$ 0.0465 |
| GBISC                                                      | UniGC (Ours) | 30     | 0.9237 $\pm$ 0.0229   | 0.8919 $\pm$ 0.0132   | 0.9074 $\pm$ 0.0089 | 0.9375 $\pm$ 0.0033 | 0.8197 $\pm$ 0.0060 | 0.1295 $\pm$ 0.0262 | 0.8749 $\pm$ 0.0409 | 0.4038 $\pm$ 0.0144 |

For whole-image berry counting, UniGC reduced MAE and RMSE relative to PMGDC [9] by 29.89% and 27.27% on WGISD and by 58.83% and 55.99% on GBISC, respectively. Its berry-point branch also achieved higher precision and  $F_1$  scores and lower pixel and normalized localization errors on both subsets. The pixel localization error was reduced by 30.06% on WGISD and 28.94% on GBISC, while the normalized error was reduced by 30.58% and 28.91%, respectively. PMGDC retained slightly higher point recall on both subsets.

For grape-cluster detection and BpGC counting, UniGC achieved higher  $F_1$ , AP50, and AP50:95 values and lower MRD and ACP-MRD on both subsets than PMGDC [9]. On WGISD, UniGC reduced MRD and ACP-MRD by 27.45% and 4.22%, respectively; on GBISC, the corresponding reductions were 9.94% and 18.37%. PMGDC achieved slightly higher detection recall on WGISD, whereas UniGC achieved higher detection precision,  $F_1$ , and AP values. On GBISC, UniGC achieved higher values for all reported detection metrics. Overall, the source-stratified results show consistent improvements in most metrics, although the magnitude of the gains varies between the two data sources.

#### 4.4. Ablation Experiments

This section presents ablation experiments and sensitivity analyses for the key modules and the main reported configuration of UniGC, sequentially examining the effects of CSD-Block, YOLO detectors, SAM mask generation models, and the BDMPA relative adsorption threshold  $\rho$  on the experimental results.

##### 4.4.1. Component Ablation of CSD-Block and Gains for the Berry-Level Backbone

To validate the contribution of CSD-Block to the berry-level counting branch and analyze the roles of its internal components, this study uses P2PNet [32] as the baseline and inserts CSD-Block or its ablated variants after the Feature Pyramid Network (FPN) output features and before the regression and classification heads. In addition to the original P2PNet, this study defines five controlled variants: Plain DWConv Block retains only standard depthwise convolution enhancement to exclude the effect introduced by additional convolution modules; CSD without Decoupling removes the explicit decoupling between the low-frequency condition component and the structural residual component; CSD without Instance Normalization (IN) removes Instance Normalization on the structural component; CSD without Multi-Dilation retains only a single dilation-rate branch; and CSD without Condition Re-coupling removes Feature-wise Linear Modulation (FiLM) and spatial-gate-based condition recoupling. All variants are compared under the same training settings.

As shown in Table 7, the full CSD-Block achieved the lowest MAE and RMSE among all evaluated variants. Relative to the original P2PNet [32], it reduced MAE from 27.7 to 23.3 and RMSE from 37.7 to 33.2, corresponding to absolute reductions of 4.4 and 4.5, respectively. By comparison, the Plain DWConv Block achieved an MAE of 27.0 and an RMSE of 37.6, remaining close to the original P2PNet. This comparison indicates that the improvement cannot be attributed solely to the addition of convolutional enhancement.

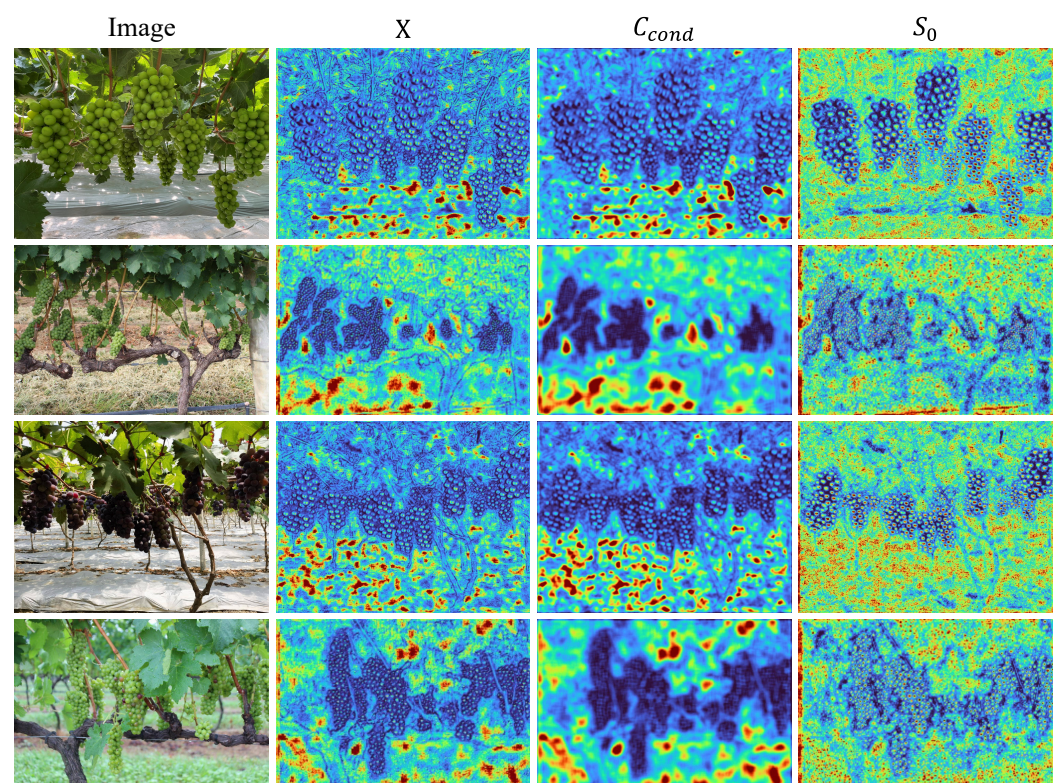
The component ablations further clarify the contribution of each design element. Removing condition–structure decoupling increased MAE and RMSE to 25.2 and 35.7, respectively, supporting the role of explicit component separation in reducing interference from smooth background responses and broader contextual variations. Removing Instance Normalization resulted in an MAE of 25.2 and an RMSE of 34.0, indicating that normalization of the structural residual contributes to stable feature enhancement. Retaining only a single dilation branch increased the errors to 24.2 and 33.4, demonstrating the benefit of aggregating structural information across multiple receptive fields. Removing condition

recoupling increased the errors to 24.6 and 34.4, suggesting that channel modulation and spatial gating facilitate the reintegration of enhanced structural information into the original feature representation.

**Table 7.** Component ablation results of the Condition-Structure Decoupled Frequency Enhancement Block (CSD-Block). Lower MAE and RMSE are better.

| Backbone Model or Variant          | MAE  | RMSE |
|------------------------------------|------|------|
| P2PNet [32]                        | 27.7 | 37.7 |
| Plain DWConv Block                 | 27.0 | 37.6 |
| CSD without Decoupling             | 25.2 | 35.7 |
| CSD without Multi-Dilation         | 24.2 | 33.4 |
| CSD without Instance Normalization | 25.2 | 34.0 |
| CSD without Condition Re-coupling  | 24.6 | 34.4 |
| Full CSD-Block                     | 23.3 | 33.2 |

Although these ablations quantify the contribution of the individual components, they do not directly show how the learned decomposition is reflected in the intermediate representations. Representative CSD-P2PNet features are therefore visualized in Figure 5.



**Figure 5.** Visualization of representative intermediate CSD-Block representations. From left to right: the input image, input feature  $X$ , learned low-pass condition component  $C_{\text{cond}}$ , and structural residual  $S_0 = X - C_{\text{cond}}$ .

Compared with the input feature  $X$ ,  $C_{\text{cond}}$  exhibits smoother spatial responses and attenuated fine-scale variations, whereas  $S_0$  preserves sharper local contrast, berry contours, and responses in densely distributed berry regions. This complementary behavior is observed across representative scenes containing different grape varieties and backgrounds. The intermediate representations are therefore consistent with the intended frequency-inspired decomposition:  $C_{\text{cond}}$  predominantly retains low-frequency contextual information, while  $S_0$  emphasizes structure-sensitive residual information.

Taken together, the component-ablation results and intermediate-feature visualizations provide complementary quantitative and qualitative evidence for the CSD-Block design. The observed improvement is supported by the combined effects of low-frequency contextual decomposition, residual structural enhancement, multi-scale feature aggregation, and condition-guided feature recoupling.

#### 4.4.2. Effect of Detector Selection on BpGC Counting

Under fixed settings of  $\rho = 0.03$ , SAM3 [46], and IoU = 0.6, Table 8 compares different YOLO detectors in terms of grape-cluster detection, point-to-cluster matching, BpGC counting, and computational efficiency. The rankings of the detector-level and downstream BpGC metrics were not fully aligned. YOLO12-L [37] achieved the highest recall, mAP@0.5, and mAP@0.5:0.95, and tied with YOLO11-S [47] for the highest  $N_{\text{match}}$  and  $R_{\text{match}}$ , but did not yield the lowest BpGC counting errors. YOLO8-L [48] obtained the lowest matched-cluster MRD, whereas YOLO26-M [49] achieved the highest precision,  $F_1$ , and  $1 - \text{FVU}$ ; however, neither detector achieved the lowest ACP-MRD.

YOLO12-S [37] achieved the lowest ACP-MRD of 0.4977 while maintaining a precision of 0.879, a recall of 0.809, an  $F_1$  score of 0.843, 996 matched grape clusters, and a matching rate of 0.792. Its matched-cluster MRD and  $1 - \text{FVU}$  were 0.1336 and 0.8557, respectively. In terms of computational efficiency, YOLO12-S contained 9.254 million parameters, required 55.10 GFLOPs at an input resolution of  $1024 \times 1024$ , and achieved 83.60 FPS in detector-only inference. Although it did not achieve the best result for every individual metric, YOLO12-S provided a favorable balance among detection accuracy, matching coverage, overall BpGC error, and computational cost. It was therefore selected as the grape-cluster detector in the main UniGC configuration.

**Table 8.** Comparison of grape-cluster detection, BpGC counting, and computational efficiency across different YOLO detectors.

| Detector      | Grape-Cluster Detection |       |       |          |               | Matching           |                    | BpGC Counting |                  |         | Computational Efficiency |        |        |
|---------------|-------------------------|-------|-------|----------|---------------|--------------------|--------------------|---------------|------------------|---------|--------------------------|--------|--------|
|               | Prec.                   | Rec.  | $F_1$ | mAP @0.5 | mAP @0.5:0.95 | $N_{\text{match}}$ | $R_{\text{match}}$ | MRD           | $1 - \text{FVU}$ | ACP-MRD | Params (M)               | GFLOPs | FPS    |
| YOLO8-N [48]  | 0.877                   | 0.780 | 0.826 | 0.863    | 0.623         | 960                | 0.763              | 0.1452        | 0.8355           | 0.5846  | 3.011                    | 20.98  | 86.60  |
| YOLO8-S [48]  | 0.895                   | 0.786 | 0.837 | 0.872    | 0.639         | 971                | 0.772              | 0.1310        | 0.8406           | 0.5256  | 11.136                   | 73.34  | 151.53 |
| YOLO8-M [48]  | 0.888                   | 0.776 | 0.828 | 0.872    | 0.648         | 963                | 0.766              | 0.1398        | 0.7903           | 0.5458  | 25.857                   | 202.41 | 74.48  |
| YOLO8-L [48]  | 0.891                   | 0.783 | 0.833 | 0.874    | 0.666         | 959                | 0.762              | 0.1279        | 0.8586           | 0.5315  | 43.631                   | 423.43 | 50.88  |
| YOLO11-N [47] | 0.884                   | 0.796 | 0.838 | 0.883    | 0.626         | 991                | 0.788              | 0.1546        | 0.8104           | 0.5757  | 2.590                    | 16.49  | 85.04  |
| YOLO11-S [47] | 0.869                   | 0.815 | 0.841 | 0.896    | 0.657         | 1020               | 0.811              | 0.1585        | 0.7927           | 0.5601  | 9.428                    | 55.16  | 105.25 |
| YOLO11-M [47] | 0.874                   | 0.804 | 0.837 | 0.882    | 0.671         | 996                | 0.792              | 0.1404        | 0.8421           | 0.5229  | 20.054                   | 174.56 | 68.37  |
| YOLO11-L [47] | 0.882                   | 0.804 | 0.841 | 0.880    | 0.670         | 976                | 0.776              | 0.1397        | 0.8481           | 0.5233  | 25.311                   | 223.42 | 60.53  |
| YOLO12-N [37] | 0.876                   | 0.801 | 0.837 | 0.875    | 0.627         | 990                | 0.787              | 0.1492        | 0.8003           | 0.5880  | 2.568                    | 16.59  | 114.71 |
| YOLO12-S [37] | 0.879                   | 0.809 | 0.843 | 0.880    | 0.656         | 996                | 0.792              | 0.1336        | 0.8557           | 0.4977  | 9.254                    | 55.10  | 83.60  |
| YOLO12-M [37] | 0.842                   | 0.820 | 0.831 | 0.889    | 0.653         | 1006               | 0.800              | 0.1481        | 0.7995           | 0.5596  | 20.138                   | 173.41 | 54.04  |
| YOLO12-L [37] | 0.866                   | 0.823 | 0.844 | 0.897    | 0.677         | 1020               | 0.811              | 0.1497        | 0.8096           | 0.5466  | 26.390                   | 228.89 | 37.96  |
| YOLO26-N [49] | 0.855                   | 0.793 | 0.823 | 0.851    | 0.622         | 938                | 0.746              | 0.1437        | 0.8035           | 0.5380  | 2.504                    | 14.78  | 134.54 |
| YOLO26-S [49] | 0.882                   | 0.792 | 0.835 | 0.872    | 0.655         | 960                | 0.763              | 0.1348        | 0.8406           | 0.5194  | 9.949                    | 57.60  | 83.89  |
| YOLO26-M [49] | 0.901                   | 0.795 | 0.845 | 0.875    | 0.666         | 966                | 0.768              | 0.1329        | 0.8665           | 0.4996  | 21.774                   | 191.28 | 73.05  |
| YOLO26-L [49] | 0.882                   | 0.793 | 0.835 | 0.869    | 0.670         | 953                | 0.758              | 0.1283        | 0.8534           | 0.5053  | 26.178                   | 238.38 | 64.87  |

#### 4.4.3. Effect of Mask Generation Model Selection on BpGC Counting

Under fixed settings of YOLO12-S [37],  $\rho = 0.03$ , and IoU = 0.6, Table 9 compares the downstream BpGC performance and computational efficiency of different SAM-family mask generators. SAM3 [46] achieved the lowest MRD and ACP-MRD, with values of 0.1336 and 0.4977, respectively, whereas SAM2.1-B [50] achieved the highest  $1 - \text{FVU}$  of 0.8645. MobileSAM [51] provided the lowest computational cost, with 10.130 million parameters, 126.93 GFLOPs per image, and 10.750 FPS.

**Table 9.** Comparison of BpGC counting and computational efficiency across SAM-family mask generators.

| SAM Model      | BpGC Counting |         |         | Computational Efficiency |              |        |
|----------------|---------------|---------|---------|--------------------------|--------------|--------|
|                | MRD           | 1 – FVU | ACP-MRD | Params (M)               | GFLOPs/Image | FPS    |
| SAM3 [46]      | 0.1336        | 0.8557  | 0.4977  | 473.585                  | 7149.94      | 3.388  |
| SAM2-B [50]    | 0.1350        | 0.8609  | 0.4987  | 80.834                   | 696.59       | 2.789  |
| SAM2.1-B [50]  | 0.1353        | 0.8645  | 0.4988  | 80.850                   | 696.59       | 3.388  |
| SAM2.1-L [50]  | 0.1397        | 0.8588  | 0.5027  | 224.447                  | 1872.66      | 2.708  |
| SAM-L [33]     | 0.1423        | 0.8446  | 0.5047  | 312.343                  | 3035.07      | 5.056  |
| SAM2-L [50]    | 0.1434        | 0.8550  | 0.5054  | 224.430                  | 1872.66      | 3.020  |
| SAM2.1-T [50]  | 0.1463        | 0.8274  | 0.5079  | 38.962                   | 342.24       | 2.940  |
| SAM2-T [50]    | 0.1490        | 0.8273  | 0.5095  | 38.946                   | 342.24       | 2.758  |
| SAM2.1-S [50]  | 0.1497        | 0.8222  | 0.5095  | 46.060                   | 409.91       | 3.078  |
| SAM2-S [50]    | 0.1497        | 0.8303  | 0.5101  | 46.044                   | 409.91       | 2.756  |
| SAM-B [33]     | 0.1525        | 0.8109  | 0.5121  | 93.735                   | 1021.47      | 7.523  |
| MobileSAM [51] | 0.1591        | 0.8070  | 0.5180  | 10.130                   | 126.93       | 10.750 |

To directly assess the generated grape-cluster masks, Table 10 reports mask quality under ground-truth-box prompts and fixed YOLO12-S predicted-box prompts matched to ground-truth clusters.

**Table 10.** Direct mask quality of SAM-family mask generators under ground-truth and predicted box prompts.

| SAM Model      | Ground-Truth-Box Prompt |                | Predicted-Box Prompt |                |                |              |
|----------------|-------------------------|----------------|----------------------|----------------|----------------|--------------|
|                | mIoU                    | Boundary $F_1$ | mIoU                 | Boundary $F_1$ | Area Log Error | Failure Rate |
| SAM3 [46]      | 0.8274                  | 0.8523         | 0.8445               | 0.8540         | 0.1003         | 0.0161       |
| SAM2-B [50]    | 0.8208                  | 0.8507         | 0.8355               | 0.8494         | 0.1019         | 0.0181       |
| SAM2.1-B [50]  | 0.8278                  | 0.8520         | 0.8437               | 0.8492         | 0.0989         | 0.0171       |
| SAM2.1-L [50]  | 0.8247                  | 0.8505         | 0.8419               | 0.8499         | 0.1022         | 0.0171       |
| SAM-L [33]     | 0.8120                  | 0.8415         | 0.8351               | 0.8464         | 0.1111         | 0.0241       |
| SAM2-L [50]    | 0.8232                  | 0.8463         | 0.8389               | 0.8465         | 0.1043         | 0.0211       |
| SAM2.1-T [50]  | 0.8009                  | 0.8160         | 0.8150               | 0.8111         | 0.1251         | 0.0361       |
| SAM2-T [50]    | 0.8004                  | 0.8187         | 0.8166               | 0.8162         | 0.1274         | 0.0321       |
| SAM2.1-S [50]  | 0.7985                  | 0.8233         | 0.8203               | 0.8249         | 0.1284         | 0.0321       |
| SAM2-S [50]    | 0.7980                  | 0.8230         | 0.8202               | 0.8247         | 0.1274         | 0.0271       |
| SAM-B [33]     | 0.7984                  | 0.8278         | 0.8201               | 0.8288         | 0.1253         | 0.0422       |
| MobileSAM [51] | 0.7670                  | 0.7832         | 0.7955               | 0.7973         | 0.1347         | 0.0442       |

Under ground-truth-box prompts, SAM2.1-B [50] achieved the highest mIoU of 0.8278, while SAM3 [46] achieved the highest boundary  $F_1$  of 0.8523. Under predicted-box prompts, SAM3 achieved the highest mIoU and boundary  $F_1$ , with values of 0.8445 and 0.8540, respectively, and the lowest failure rate of 0.0161. SAM2.1-B obtained the lowest area log error of 0.0989.

These results show that SAM3 combines favorable mask quality under the box prompts used in the UniGC pipeline with the lowest downstream MRD and ACP-MRD. It was therefore retained as the accuracy-oriented mask generator in the main configuration. However, its substantially higher computational cost indicates a clear accuracy–efficiency trade-off, and the observed downstream differences may also reflect interactions between the generated masks and BDMPA.

#### 4.4.4. Sensitivity Analysis of the BDMPA Relative Adsorption Threshold $\rho$

Under fixed settings of YOLO12-S, SAM3, and IoU = 0.6, Table 11 evaluates the sensitivity of BpGC counting to the relative adsorption threshold  $\rho$ . Increasing  $\rho$  from 0 to 0.03 reduced MRD from 0.1424 to 0.1336 and ACP-MRD from 0.5046 to 0.4977, while increasing  $1 - \text{FVU}$  from 0.8420 to 0.8557. This trend indicates that limited external adsorption can recover near-boundary berry points that are not covered by locally incomplete masks. When  $\rho$  exceeded 0.03, MRD and ACP-MRD no longer improved, whereas  $1 - \text{FVU}$  varied only marginally and reached its maximum of 0.8558 at  $\rho = 0.05$ . Therefore,  $\rho = 0.03$  was selected because it jointly minimized MRD and ACP-MRD while maintaining  $1 - \text{FVU}$  close to its maximum.

**Table 11.** BpGC counting results under different  $\rho$  settings.

| $\rho$ | MRD    | $1 - \text{FVU}$ | ACP-MRD |
|--------|--------|------------------|---------|
| 0.00   | 0.1424 | 0.8420           | 0.5046  |
| 0.01   | 0.1387 | 0.8487           | 0.5017  |
| 0.02   | 0.1361 | 0.8527           | 0.4996  |
| 0.03   | 0.1336 | 0.8557           | 0.4977  |
| 0.04   | 0.1346 | 0.8556           | 0.4984  |
| 0.05   | 0.1345 | 0.8558           | 0.4984  |
| 0.06   | 0.1349 | 0.8554           | 0.4987  |
| 0.07   | 0.1353 | 0.8552           | 0.4990  |
| 0.08   | 0.1362 | 0.8544           | 0.4997  |
| 0.09   | 0.1362 | 0.8541           | 0.4997  |
| 0.10   | 0.1365 | 0.8546           | 0.5000  |
| 0.15   | 0.1386 | 0.8521           | 0.5016  |
| 0.20   | 0.1400 | 0.8489           | 0.5027  |

#### 4.4.5. Ablation Analysis of BDMPA Point-Cluster Assignment Strategies

To validate the roles of internal boundary-depth disambiguation and external adsorption in BDMPA, this study compares six point-cluster assignment strategies under fixed settings of YOLO12-S [37], SAM3 [46], and IoU = 0.6. All strategies use the same predicted berry points, grape-cluster detection boxes, and SAM-generated masks; therefore, the comparison focuses on the effect of the assignment strategy itself on BpGC counting.

The compared strategies are defined as follows. Box-only assigns a point to a YOLO detection box if the point falls inside the box; when multiple boxes contain the same point, the box with the highest detection confidence is selected, and points outside all boxes are discarded. Mask-only assigns a point to a SAM mask if the point falls inside the mask; when multiple masks contain the same point, the mask associated with the highest-confidence detection box is selected, and uncovered points are discarded. Nearest-mask assigns each point to the nearest SAM mask according to the Euclidean distance to the mask boundary, without using the  $\rho$  constraint. Mask + centroid tie-break assigns only mask-covered points and resolves overlapping-mask ambiguity by selecting the mask whose centroid is closest to the point. BDMPA without external adsorption uses normalized internal boundary-depth disambiguation for mask-covered points, while points outside all masks are discarded. Full BDMPA further introduces  $\rho$ -constrained external adsorption after internal boundary-depth disambiguation.

As shown in Table 12, Full BDMPA achieved the lowest MRD and ACP-MRD, with values of 0.1336 and 0.4977, respectively, together with the highest  $1 - \text{FVU}$  of 0.8557. Box-only provided useful coarse spatial constraints but remained limited by rectangular boundaries, whereas Mask-only and Nearest-mask produced larger MRD and ACP-MRD values, indicat-

ing that mask containment alone or unconstrained nearest-mask assignment is insufficient for reliable point-to-cluster association.

Replacing centroid-based ambiguity resolution with normalized internal boundary depth reduced MRD from 0.1451 to 0.1424 and ACP-MRD from 0.5068 to 0.5046. Introducing  $\rho$ -constrained external adsorption further reduced MRD and ACP-MRD to 0.1336 and 0.4977, respectively, while increasing  $1 - \text{FVU}$  from 0.8420 to 0.8557. Relative to BDMPA without external adsorption, Full BDMPA reduced MRD by 6.18% and ACP-MRD by 1.37%. These results support the complementary roles of internal boundary-depth disambiguation and constrained external adsorption in improving both matched-cluster counting and overall BpGC performance.

**Table 12.** Ablation results of different point-to-cluster assignment strategies for BpGC counting.

| Assignment Strategy               | MRD    | $1 - \text{FVU}$ | ACP-MRD |
|-----------------------------------|--------|------------------|---------|
| Box-only                          | 0.1448 | 0.8432           | 0.5065  |
| Mask-only                         | 0.1642 | 0.7807           | 0.5219  |
| Nearest-mask                      | 0.1735 | 0.7517           | 0.5293  |
| Mask + centroid tie-break         | 0.1451 | 0.8364           | 0.5068  |
| BDMPA without external adsorption | 0.1424 | 0.8420           | 0.5046  |
| Full BDMPA                        | 0.1336 | 0.8557           | 0.4977  |

#### 4.4.6. Pipeline-Level Ablation of Berry-Point Prediction and Point-to-Cluster Assignment

To disentangle the contributions of the berry-point branch and the point-to-cluster assignment strategy, four controlled pipeline configurations were evaluated over three runs with different random seeds. All configurations used the same YOLO12-S detector [37] and SAM3 mask generator [46], while varying only the berry-point branch and assignment strategy. Table 13 summarizes the results.

**Table 13.** Three-run pipeline-level ablation of the berry-point branch and assignment strategy for BpGC counting.

| Pipeline Configuration             | MRD                 | $1 - \text{FVU}$    | ACP-MRD             |
|------------------------------------|---------------------|---------------------|---------------------|
| P2PNet [32] + Mask-Only Assignment | $0.1637 \pm 0.0034$ | $0.7876 \pm 0.0258$ | $0.5334 \pm 0.0149$ |
| CSD-P2PNet + Mask-Only Assignment  | $0.1572 \pm 0.0064$ | $0.7929 \pm 0.0262$ | $0.5283 \pm 0.0119$ |
| P2PNet [32] + Full BDMPA           | $0.1391 \pm 0.0057$ | $0.8494 \pm 0.0159$ | $0.5213 \pm 0.0120$ |
| UniGC (Ours)                       | $0.1288 \pm 0.0043$ | $0.8589 \pm 0.0098$ | $0.5062 \pm 0.0124$ |

With mask-only assignment, replacing P2PNet [32] with CSD-P2PNet reduced the matched-cluster MRD from  $0.1637 \pm 0.0034$  to  $0.1572 \pm 0.0064$  and ACP-MRD from  $0.5334 \pm 0.0149$  to  $0.5283 \pm 0.0119$ . With Full BDMPA, the same replacement further reduced the matched-cluster MRD from  $0.1391 \pm 0.0057$  to  $0.1288 \pm 0.0043$  and ACP-MRD from  $0.5213 \pm 0.0120$  to  $0.5062 \pm 0.0124$ . Conversely, replacing mask-only assignment with Full BDMPA improved all three BpGC metrics for both berry-point branches.

The full UniGC pipeline achieved the lowest matched-cluster MRD and ACP-MRD and the highest  $1 - \text{FVU}$ . Relative to the P2PNet [32] with mask-only assignment configuration, UniGC reduced the matched-cluster MRD and ACP-MRD by 21.32% and 5.10%, respectively, while increasing  $1 - \text{FVU}$  by 9.05%. These controlled comparisons suggest that CSD-P2PNet and Full BDMPA provide complementary benefits for BpGC counting.

#### 4.4.7. Oracle-Based Stage-Wise Error Decomposition

To quantify stage-wise error propagation in UniGC, three single-stage oracle diagnostics were evaluated by replacing the berry-point localization, grape-cluster detection

and prompt geometry, or cluster-mask output with its corresponding oracle output while retaining the remaining predicted stages. Table 14 summarizes the results over three runs, and  $\Delta$ ACP-MRD denotes the absolute reduction in mean ACP-MRD relative to the Full UniGC baseline. Because MRD and  $1 - \text{FVU}$  are calculated on the configuration-specific matched sets, ACP-MRD is used as the principal metric for cross-configuration comparison.

**Table 14.** Oracle-based decomposition of stage-wise errors in the UniGC pipeline over three runs.

| Configuration                        | MRD                 | $1 - \text{FVU}$    | ACP-MRD             | $\Delta$ ACP-MRD |
|--------------------------------------|---------------------|---------------------|---------------------|------------------|
| Full UniGC                           | $0.1288 \pm 0.0043$ | $0.8589 \pm 0.0098$ | $0.5062 \pm 0.0124$ | —                |
| Oracle Berry-Point Localization      | $0.0839 \pm 0.0023$ | $0.8875 \pm 0.0106$ | $0.4783 \pm 0.0099$ | 0.0279           |
| Oracle Detection and Prompt Geometry | $0.1289 \pm 0.0021$ | $0.9375 \pm 0.0022$ | $0.1289 \pm 0.0021$ | 0.3773           |
| Oracle Cluster Masks                 | $0.0935 \pm 0.0047$ | $0.9263 \pm 0.0136$ | $0.4858 \pm 0.0059$ | 0.0204           |

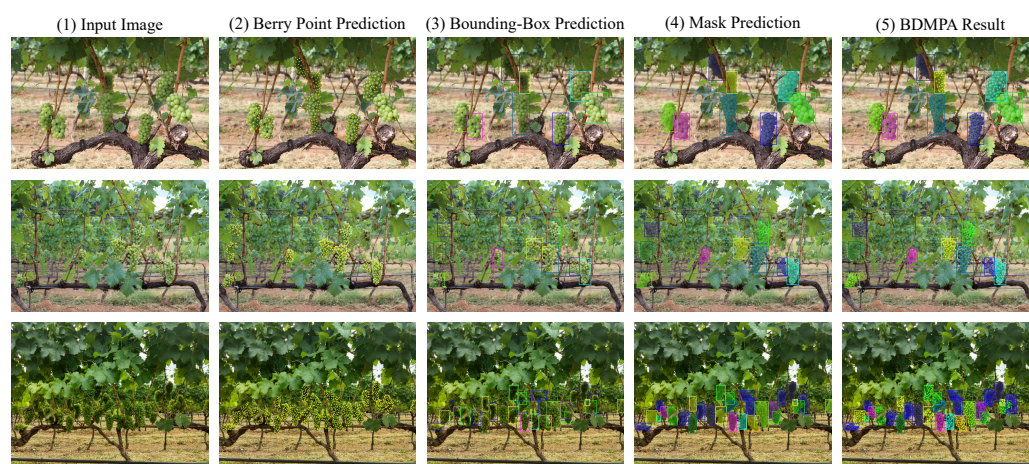
Among the three stage replacements, oracle detection and prompt geometry produced the largest reduction in ACP-MRD, decreasing it from  $0.5062 \pm 0.0124$  to  $0.1289 \pm 0.0021$ , with an absolute mean reduction of 0.3773. Oracle berry-point localization and oracle cluster masks reduced ACP-MRD to  $0.4783 \pm 0.0099$  and  $0.4858 \pm 0.0059$ , corresponding to smaller absolute reductions of 0.0279 and 0.0204, respectively. Under the evaluated oracle settings, grape-cluster detection availability and the resulting prompt geometry therefore contributed most strongly to the all-cluster penalized error.

The oracle results also show that berry-point localization and mask generation remain measurable sources of downstream error, although their effects on ACP-MRD were smaller than that of detection and prompt geometry. These findings indicate that improving the completeness and geometric accuracy of upstream grape-cluster detection is likely to provide the largest improvement in the current pipeline, while further refinement of berry localization and cluster masks may provide additional gains.

#### 4.5. Qualitative Visualization

To complement the quantitative ablation of point-to-cluster assignment strategies, Figure 6 visualizes the UniGC inference process from multi-level prediction to final point-to-cluster assignment. From left to right, the figure shows the input image, berry-point predictions generated by CSD-P2PNet, grape-cluster bounding boxes detected by YOLO12-S, cluster instance masks generated by SAM3 under box prompts, and the final assignment results produced by BDMPA. This visualization illustrates how berry-point predictions and cluster-level geometric priors are integrated to produce cluster-wise berry counts.

As shown in Figure 6, the berry-point branch provides dense point-level localization results, whereas the detection and mask-generation stages provide coarse-to-fine cluster-level spatial priors. After applying BDMPA, berry points are assigned to corresponding grape cluster instances in a per-point manner. The visualization also shows typical cases near cluster boundaries, where simple in-box or in-mask assignment may be affected by overlapping clusters, locally incomplete masks, or boundary jitter. By combining internal boundary-depth disambiguation with  $\rho$ -constrained external adsorption, BDMPA produces more coherent point-to-cluster assignment results, which is consistent with the quantitative gains reported in Table 12.



**Figure 6.** Qualitative visualization of the UniGC inference process. From left to right: input image, berry-point predictions, grape-cluster bounding-box predictions, SAM-generated cluster masks, and final BDMPA point-to-cluster assignment results.

#### 4.6. Computational Efficiency of the Complete UniGC Pipeline

The computational complexity and runtime efficiency of the complete UniGC pipeline were evaluated on the hardware described in Section 4.1 using a batch size of 1. Pure-computation latency includes the CSD-P2PNet, YOLO12-S, SAM3, and BDMPA stages, whereas application-level wall-clock time additionally includes the surrounding data-handling overhead. Because BDMPA is a non-neural geometric procedure, a unified GFLOPs estimate is not reported for the complete pipeline. Stage-wise latency was measured within the complete pipeline and is therefore not directly comparable with the standalone throughput results reported for individual YOLO and SAM variants. The results are summarized in Table 15.

**Table 15.** Computational complexity and runtime efficiency of the complete UniGC pipeline.

| Panel A. Parameter and Computational Complexity |                          |             |                      |       |                                      |
|-------------------------------------------------|--------------------------|-------------|----------------------|-------|--------------------------------------|
| Component                                       | Total Parameters         |             | Optimized Parameters |       | GFLOPs                               |
| CSD-P2PNet                                      | 22,432,404               |             | 22,432,404           |       | 881.8190                             |
| YOLO12-S                                        | 9,253,523                |             | 9,253,507            |       | 55.0965                              |
| SAM3                                            | 473,585,282              |             | 0                    |       | 7149.9378/image                      |
| BDMPA                                           | 0                        |             | 0                    |       |                                      |
| Complete UniGC                                  | 505,271,209              |             | 31,685,911           |       | N/A                                  |
| Panel B. Stage-Wise Pure-Computational Latency  |                          |             |                      |       |                                      |
| Stage                                           | Mean Latency (ms)        |             | Percentage of Total  |       |                                      |
| CSD-P2PNet                                      | 149.304                  |             | 17.2%                |       |                                      |
| YOLO12-S                                        | 43.017                   |             | 4.9%                 |       |                                      |
| SAM3 Image Encoder                              | 243.389                  |             | 28.0%                |       |                                      |
| SAM3 Prompt/Mask Decoder                        | 96.735                   |             | 11.1%                |       |                                      |
| BDMPA                                           | 337.027                  |             | 38.8%                |       |                                      |
| Total                                           | 869.472                  |             | 100.0%               |       |                                      |
| Panel C. Complete-Pipeline Runtime and Memory   |                          |             |                      |       |                                      |
| Timing Scope                                    | Mean $\pm$ SD (ms/Image) | Median (ms) | P95 (ms)             | FPS   | Peak Allocated/Reserved Memory (MiB) |
| Pure Computation                                | 869.472 $\pm$ 411.277    | 687.619     | 1729.733             | 1.150 | 8533.2/17,554.0                      |
| Application-Level Wall Clock                    | 1009.629 $\pm$ 511.687   | 760.807     | 2048.668             | 0.990 | —                                    |

The complete pipeline comprises 505.27 million parameters, of which 31.69 million are optimized during training. SAM3 is used without fine-tuning, and BDMPA introduces no trainable parameters. Pure-computation time averaged  $869.472 \pm 411.277$  ms per image, corresponding to 1.150 FPS, whereas the application-level wall-clock time was

1009.629  $\pm$  511.687 ms per image, corresponding to 0.990 FPS. The peak allocated and reserved GPU memory reached 8533.2 MiB and 17,554.0 MiB, respectively.

BDMPA accounted for 38.8% of the pure-computation latency. The SAM3 image encoder and prompt/mask decoder jointly accounted for a further 39.1%, while CSD-P2PNet and YOLO12-S contributed 17.2% and 4.9%, respectively. The current implementation is therefore more suitable for offline or server-side analysis than for real-time inference on resource-constrained onboard hardware. Future efficiency optimization should primarily target lightweight mask generation and parallel or vectorized point-to-cluster assignment.

## 5. Discussion

This study proposes UniGC, a unified framework that integrates whole-image counting of image-visible berries, grape-cluster detection and counting, and BpGC counting within a common inference pipeline. The framework combines a point-based berry branch derived from P2PNet [32], a YOLO12-S grape-cluster detector [37], box-prompted cluster-mask generation with SAM3 [46], and BDMPA-based point-to-cluster assignment. Unlike single-task approaches that provide only image-level counts or cluster locations, UniGC explicitly links image-visible berry-point localization, grape-cluster instance geometry, and cluster-wise berry counting. Its trainable branches require berry center-point annotations and grape-cluster bounding-box annotations. By producing associated outputs at the berry, cluster, and within-cluster levels, UniGC provides structured information on image-visible berry load. Compared with isolated image-level counts, these outputs may provide a more informative basis for crop-load characterization, identification of clusters with atypical visible berry loads, and cluster-specific thinning or management decisions.

Existing studies on grape cluster detection or instance segmentation can provide cluster-level geometric priors [13], but they usually cannot directly provide cluster-wise counts of image-visible berries; although berry-level density-map, probability-map, or segmentation methods can improve local counting accuracy, they still require additional post-processing to convert responses into instances or intra-cluster counts [26,27,35]. Yang et al. [9] proposed PMGDC, which models grape clusters, berries, and BpGC as related tasks and provides an important basis for unified counting; however, probability maps and multi-stage post-processing may still lead to misassignment under adjacent-cluster overlap, boundary adhesion, and response drift. Du et al. [10] proposed MVDNet, which jointly estimates intra-cluster berry counts through density-map regression and cluster-level segmentation, and also shows strong potential; however, its segmentation branch relies on polygon- or pixel-level mask supervision, which is more fine-grained than the point-and-box supervision adopted in this study, and its main evaluation metrics differ from the MRD and 1 – FVU used in this study, making a fully equivalent direct comparison difficult. Within the UniGC framework, berry predictions are explicitly represented as coordinate points, while the SAM framework proposed by Kirillov et al. [33] generates cluster-level masks from box prompts to provide instance-level geometric boundaries. BDMPA subsequently establishes point-to-cluster assignments by integrating these two forms of information. In this way, UniGC explicitly links berry-point localization and cluster membership estimation within the unified multi-level counting framework.

The repeated-run results support the principal design choices. CSD-P2PNet, built on P2PNet [32], achieved an MAE of  $23.4407 \pm 0.1577$ , an RMSE of  $32.8898 \pm 1.2682$ , and an  $F_1$  score of  $0.8374 \pm 0.0097$  for image-visible berry-point localization. YOLO12-S [37] achieved an  $F_1$  score of  $0.8433 \pm 0.0056$  and an AP50 of  $0.8789 \pm 0.0080$  for grape-cluster detection. For BpGC counting, UniGC obtained a matched-cluster MRD of  $0.1288 \pm 0.0043$ , a matched-cluster 1 – FVU of  $0.8589 \pm 0.0098$ , and an all-cluster penalized ACP-MRD of  $0.5062 \pm 0.0124$ . Relative to PMGDC [9], UniGC reduced the matched-cluster MRD

and ACP-MRD by 20.15% and 2.43%, respectively, while increasing the matched-cluster  $1 - \text{FVU}$  by 3.91%. These results support the complementary contributions of image-visible berry-point localization, grape-cluster geometry, and point-to-cluster assignment to BpGC performance within the evaluated unified framework. From an agronomic perspective, the combined outputs of whole-image visible berry load, cluster locations, and per-cluster visible berry counts provide a more informative image-based basis for crop-load characterization than any single output alone. Although these image-derived outputs do not directly estimate severely occluded or invisible berries, cluster mass, or field-scale yield, they provide a feasible visual-information basis for subsequent integration with complementary phenotypic traits, multi-view or depth information, and calibration models toward more comprehensive crop-load and yield estimation.

The contribution of BDMPPA is mainly reflected in its handling of boundary uncertainty. Simple in-box or in-mask rules assume reliable geometric boundaries, but natural orchard images often contain undersized masks, boundary jitter, multiple overlapping masks, and berry points located outside boundaries. BDMPPA adopts the rule of “mandatory internal assignment and cautious external adsorption”: points covered by multiple masks are disambiguated using normalized internal boundary depth, whereas external points are adsorbed only within the relative distance threshold. The sensitivity analysis shows that when  $\rho$  increases from 0 to 0.03, MRD decreases from 0.1424 to 0.1336, ACP-MRD decreases from 0.5046 to 0.4977, and  $1 - \text{FVU}$  increases from 0.8420 to 0.8557. When  $\rho$  exceeds 0.03, MRD and ACP-MRD no longer improve consistently, while  $1 - \text{FVU}$  changes only marginally. This indicates that limited outside-boundary adsorption can recover near-boundary berry points that are excluded by locally incomplete masks, whereas an overly wide adsorption range relaxes the geometric constraint and may increase the possibility of incorrect assignment.

Although UniGC achieves competitive results, several limitations remain. First, the unified modular formulation remains sensitive to inter-stage error propagation. The single-stage oracle diagnostics in Table 14 show that replacing grape-cluster detection and prompt geometry produced the largest reduction in ACP-MRD (0.3773), whereas replacing berry-point localization or cluster masks yielded smaller reductions of 0.0279 and 0.0204, respectively. These results identify grape-cluster detection completeness and prompt geometry as the principal bottleneck of the current pipeline, indicating that upstream localization and geometric accuracy should be prioritized. Second, the source-stratified evaluation showed broadly consistent comparative trends across WGSD and GBISC, although the magnitude of the gains varied between the two sources. Because both sources contributed to the training and evaluation splits, these results do not constitute an independent cross-domain evaluation. Furthermore, we did not identify an additional publicly available grape dataset that simultaneously provides the berry-point annotations, grape-cluster annotations, and point-to-cluster correspondence labels required by the current training and BpGC evaluation protocol. Therefore, independent external BpGC evaluation under a directly compatible protocol remains unavailable. Environmental variations may affect different stages of the unified framework in distinct but related ways. Changes in cultivar appearance, berry size and color, cluster compactness, growth stage, canopy structure, and background composition may alter local berry textures and cluster boundaries. Variations in acquisition distance, viewing angle, image resolution, sensor characteristics, season, and region may further affect berry-point localization, detection-box completeness, and prompt geometry. Illumination changes, including shadows, highlights, strong backlighting, and nighttime artificial illumination, may introduce local contrast shifts and boundary ambiguities, thereby influencing point prediction, mask generation, and subsequent point-to-cluster assignment. These errors can propagate through the unified pipeline

and affect the final BpGC results. Related cross-site visual inspection research has similarly shown that model performance may degrade under unseen acquisition conditions and that domain adaptation can mitigate such distribution shifts [52]. Third, the current model mainly counts image-visible berry points and does not directly estimate severely occluded or invisible berries or jointly model yield-related traits such as berry size, berry mass, cluster compactness, and cluster weight. Nevertheless, its whole-image visible berry counts, cluster locations, and per-cluster visible berry counts provide a structured basis for subsequent integration with multi-view imagery, depth information, physical measurements, and calibration models. The study by Parr et al. [1] further demonstrates the potential of combining low-cost RGB-D acquisition with complementary phenotypic information for crop-load and yield estimation. Fourth, the complete UniGC pipeline remains computationally intensive. On the evaluated RTX 4090D server, the application-level runtime was 1009.629 ms per image, corresponding to 0.990 FPS, while the peak allocated and reserved GPU memory reached 8533.2 MiB and 17,554.0 MiB, respectively. The current configuration is therefore more suitable for offline or server-side analysis than for real-time onboard inference. Device-level deployment experiments were not conducted because representative mobile-robot, unmanned-aerial-vehicle, and edge-computing platforms and the corresponding system-integration conditions were unavailable in the present experimental environment.

Our future work will proceed in the following four directions. First, end-to-end or weakly coupled multi-task networks will be explored under comparable point-and-box supervision to promote representation sharing among grape-cluster detection, berry-point localization, mask generation, and point-to-cluster assignment and thereby reduce inter-stage error propagation. Second, we will further construct and annotate a large-scale, multi-scene BpGC dataset containing berry points, grape-cluster annotations, and point-to-cluster correspondence labels. The dataset will progressively cover additional vineyards, cultivars, growth stages, cluster structures, acquisition distances, viewing angles, illumination conditions, and imaging devices. This dataset will enable independent external evaluation and vineyard-, cultivar-, and acquisition-separated testing, while also supporting further investigation of domain adaptation and few-shot learning for unseen vineyard environments. Third, multi-view imagery, RGB-D sensing, physical berry and cluster measurements, and calibration models will be incorporated to extend UniGC from image-visible counting toward hidden-berry estimation and complementary yield-trait analysis. The cluster locations, visible berry load, and BpGC outputs will also be connected to fruit-thinning decisions, picking-point prediction, and yield-estimation workflows. Fourth, lightweight mask generators, reusable image embeddings, and parallel or vectorized implementations of BDMPA will be investigated to reduce computational overhead. When suitable mobile-robot, unmanned-aerial-vehicle, or edge-computing platforms and the required system-integration conditions become available, device-level experiments will be conducted to evaluate inference latency, memory consumption, energy efficiency, and operational stability in practical vineyard environments.

## 6. Conclusions

This paper proposes UniGC, a unified framework for multi-level grape counting in natural orchard images. The framework integrates grape-cluster detection and counting, image-visible berry localization and counting, and BpGC counting into a unified pipeline. First, CSD-P2PNet incorporating CSD-Block is used to perform berry-point localization and whole-image visible-berry counting. Second, YOLO predicts grape-cluster bounding boxes, which are used as prompts for SAM to generate the corresponding cluster instance masks as geometric priors. Finally, BDMPA, which requires no additional trainable parameters,

assigns each predicted berry point to at most one grape-cluster instance and produces per-cluster visible berry counts.

The experimental results show that UniGC achieves stable and competitive multi-level counting performance on the mixed dataset organized from WGISD and GBISC, while its trainable branches use only berry-point annotations and grape-cluster bounding-box annotations. For whole-image berry counting and point localization, CSD-P2PNet achieves favorable performance compared with the evaluated methods. For grape-cluster detection and BpGC counting, UniGC demonstrates clear and consistent overall improvements over the representative baseline PMGDC in most evaluated metrics, particularly in grape-cluster localization quality and intra-cluster counting accuracy. Overall, UniGC provides a methodological reference for multi-level grape counting under point-and-box supervision and may inform the development of related fruit-counting and visual phenotyping methods. Future work will focus on developing end-to-end modules for multi-level counting, constructing large-scale multi-scene BpGC datasets, conducting independent cross-domain evaluation, integrating complementary phenotypic information, and improving computational efficiency to support practical applications.

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## Abbreviations

The following abbreviations are used in this manuscript:

|           |                                                                |
|-----------|----------------------------------------------------------------|
| UniGC     | Unified Multi-Level Grape Counting framework                   |
| BpGC      | Berries-per-Grape-Cluster                                      |
| CSD-Block | Condition-Structure Decoupled Frequency Enhancement Block      |
| BDMPA     | Boundary-Depth Guided Mask-Point Assignment                    |
| YOLO      | You Only Look Once                                             |
| SAM       | Segment Anything Model                                         |
| WGISD     | Wine Grape Instance Segmentation Dataset                       |
| GBISC     | Sichuan Grape Berry Instance Segmentation and Counting Dataset |
| FPN       | Feature Pyramid Network                                        |
| FiLM      | Feature-wise Linear Modulation                                 |
| MLP       | Multi-layer perceptron                                         |
| IN        | Instance Normalization                                         |
| LPF       | Low-Pass Filter                                                |
| SiLU      | Sigmoid linear unit                                            |
| DWConv    | Depthwise convolution                                          |

|            |                                               |
|------------|-----------------------------------------------|
| PWConv     | Pointwise convolution                         |
| GAP        | Global average pooling                        |
| ROI        | Region of interest                            |
| IoU        | Intersection over union                       |
| GT         | Ground truth                                  |
| MAE        | Mean absolute error                           |
| RMSE       | Root mean square error                        |
| mAP        | Mean average precision                        |
| MRD        | Mean relative deviation                       |
| FVU        | Fraction of variance unexplained              |
| CSD-P2PNet | Condition-Structure Decoupled P2PNet          |
| ACP-MRD    | All-Cluster Penalized Mean Relative Deviation |
| mIoU       | Mean intersection over union                  |
| SD         | Standard deviation                            |
| FPS        | Frames per second                             |
| GFLOPs     | Giga floating-point operations                |

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